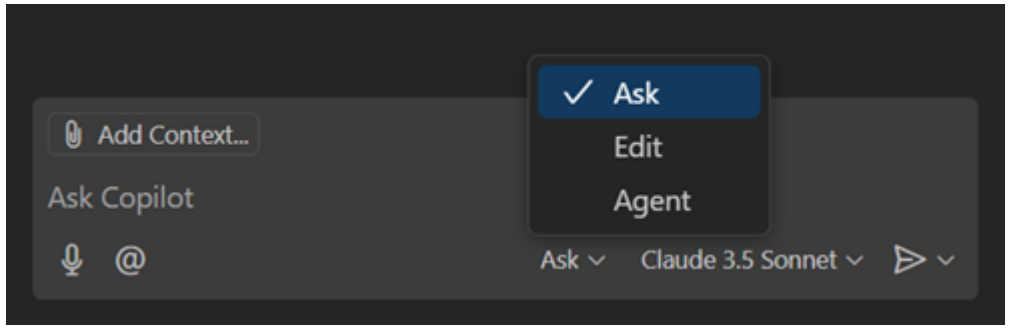
Exeircise: AI Assisted Coding

Copilot Chat modes

Ask	**Conversational Assistance** The standard chat interface where you ask questions, request explanations, or get code snippets without modifying files directly.	• Understanding existing code • Debugging concepts • Generating small snippets to copy-paste
Plan	**Architecture & Strategy** Analyzes your request and codebase to build a step-by-step implementation plan (blueprint) without writing code immediately.	• Breaking down complex features • clarifying requirements before coding • Designing large refactors
Edit	**Controlled Modifications** Allows you to select specific files and instruct Copilot to apply granular changes directly to them.	• Refactoring a specific file • Fixing a known bug in a function • Cleaning up code logic
Agent	**Autonomous Execution** An autonomous mode that can plan, edit, run terminal commands, and fix errors across multiple files to complete a task from start to finish.	• End-to-end feature implementation • Complex multi-file tasks • Tasks requiring iterative testing and fixing

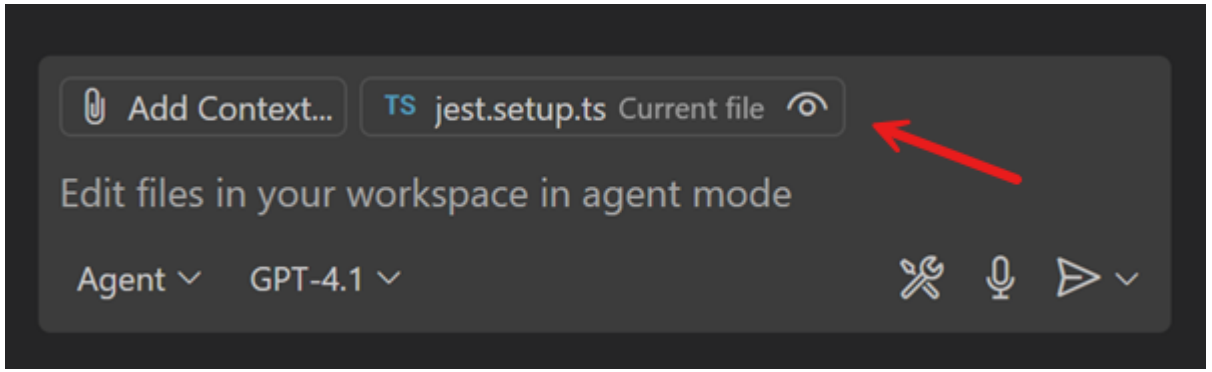
Ask Mode

- Answering your question without touching your code.
  - Explain what the code does, suggest how to test it, give you a code snippet that implements what you are asking about
  - How to use a certain library, how to structure a SQL query, etc



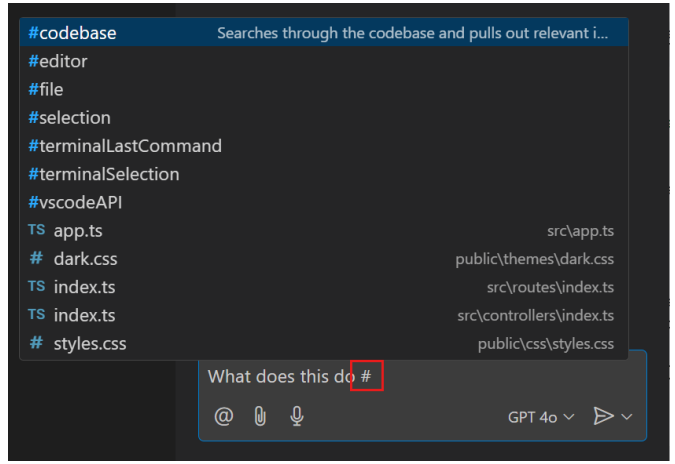
Implicit Context

- VS Code automatically adds the active file as a context item in the chat input box (indicated by Current file)

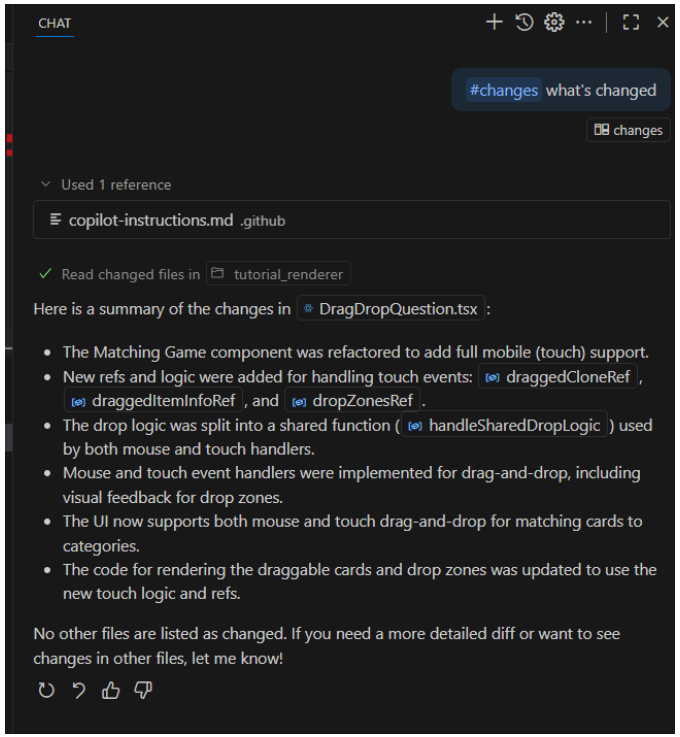


#-mentions

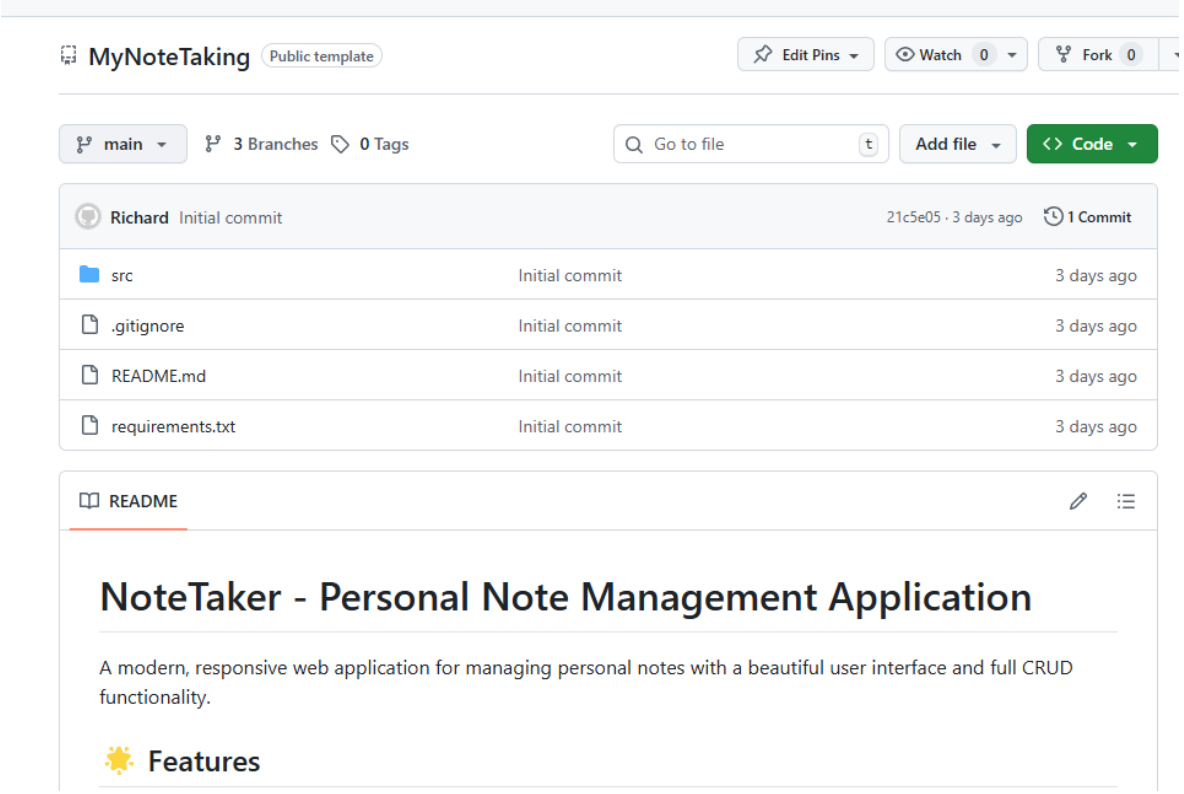
- # followed by the file name, folder name, or symbol name
- #codebase
  - perform a codebase search for your workspace.
- #fetch
  - Reference contents from the web (e.g. updated API doc)

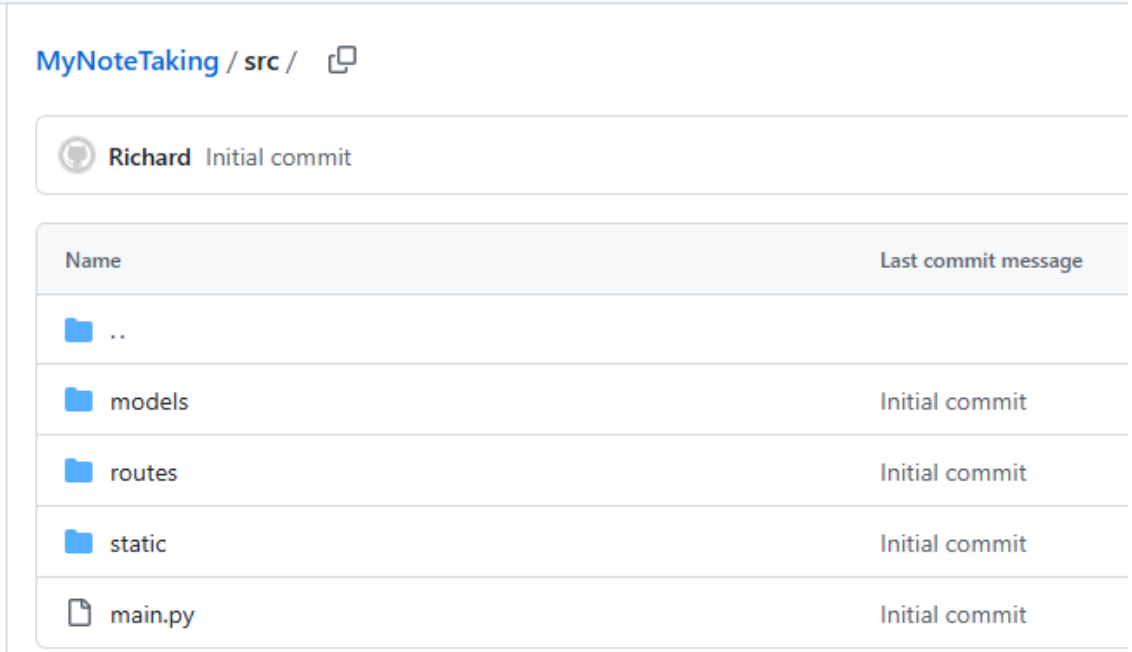


Category	Examples
Reference your pending source control changes	- "Summarize the #changes“ - "Generate release notes based on the #changes"
Understand the codebase	- "Explain how authentication works in #codebase" - "Where is the database connecting string configured? #codebase“ - How do I build this #codebase?" - "Where is #getUser used? #usages"
Generate code that is consistent with your codebase	- "Create an about page and include it in the nav bar #codebase“ - "Add a new API route for updating the address info #codebase“ - "Add a login button and style it based on #styles.css"
Fix issues in the workspace	- "Fix the issues in #problems" - "Fix the failing tests #testFailure"
Reference content from the web	- "How do I use the useState hook in react 18? #fetch https://18.react.dev/reference/react/useState#usage" - "Build an API endpoint to fetch address info, use the template from #githubRepo contoso/api-templates"



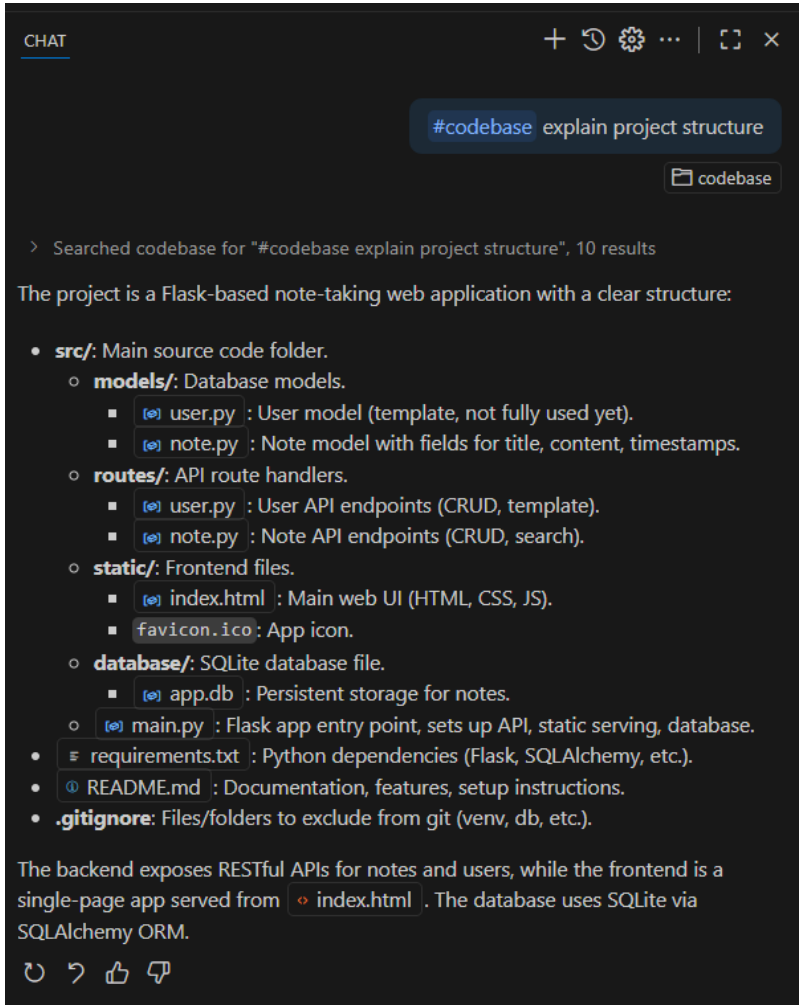
The note taking app (Generated with Manus AI)





The note taking app developed by Manus AI in lecture 1: https://3dhkilc88dkk.manus.space/

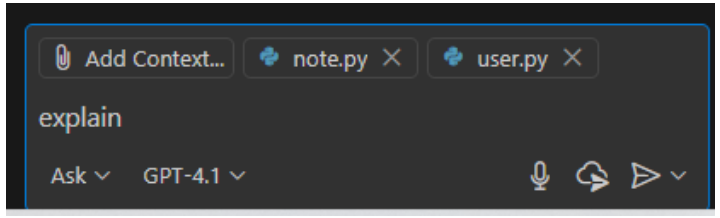
Let's explore the codebase

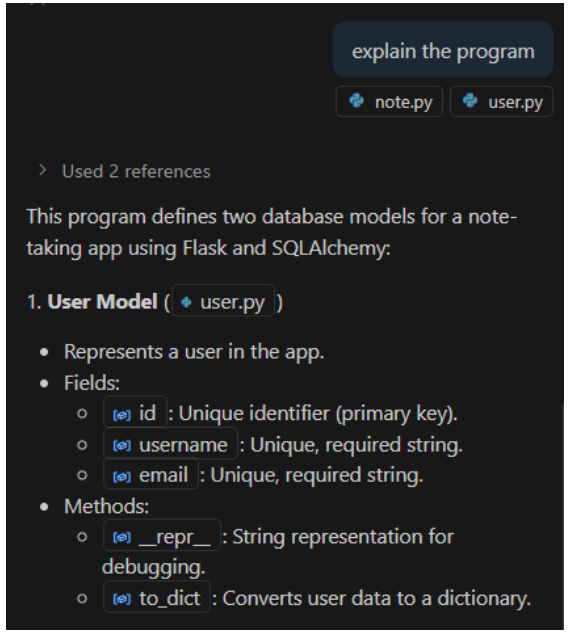


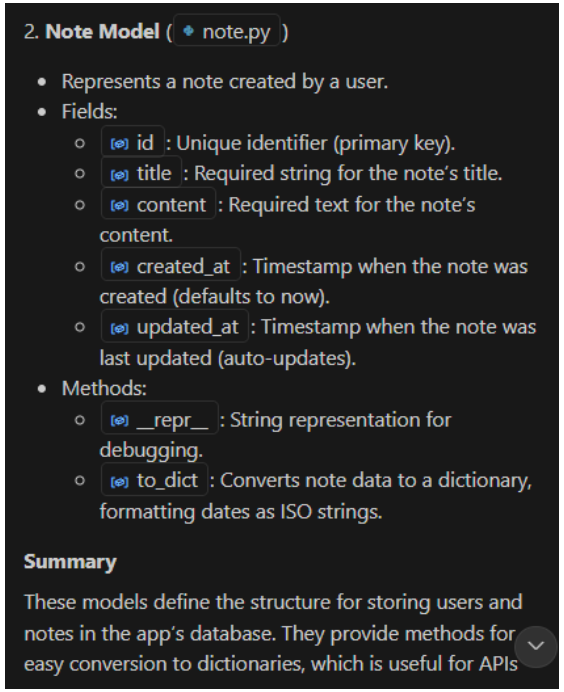
#codebase

A chat variable used in Copilot Chat that references relevant context from your entire workspace or codebase and adds that context to your chat prompt.

- Add context to ask questions







which part of the code handle creating notes

The part of the code that handles creating notes is typically found in the route handler for POST requests to the notes endpoint. In your codebase, this is likely in:

```
@note_bp.route('/notes', methods=['POST']
def create_note():
    # ...existing code...
    # Logic for creating a new note (e.g.
    # ...existing code...
```

This function receives the note data from the client, processes it, and adds the new note to your storage (such as a database).

VS

which part of the code handle creating notes #codebase

codebase

Searched codebase for "which part of the code handle creating notes #codebase ", 10 results

The part of the code that handles creating notes is in note.py , specifically the create_note function:

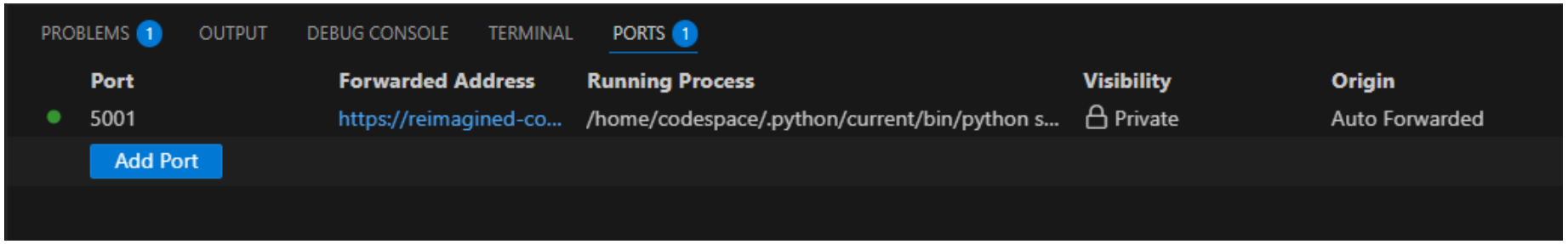
```
@note_bp.route('/notes', methods=['POST']
def create_note():
    """Create a new note"""
    try:
        data = request.json
        if not data or 'title' not in dat
            return jsonify({'error': 'Tit

        note = Note(title=data['title'],
        db.session.add(note)
        db.session.commit()
        return jsonify(note.to_dict()), 2
    except Exception as e:
        db.session.rollback()
        return jsonify({'error': str(e)})
```

This function receives a POST request to `/api/notes`, validates the input, creates a new Note object, saves it to the database, and returns the created note as JSON.

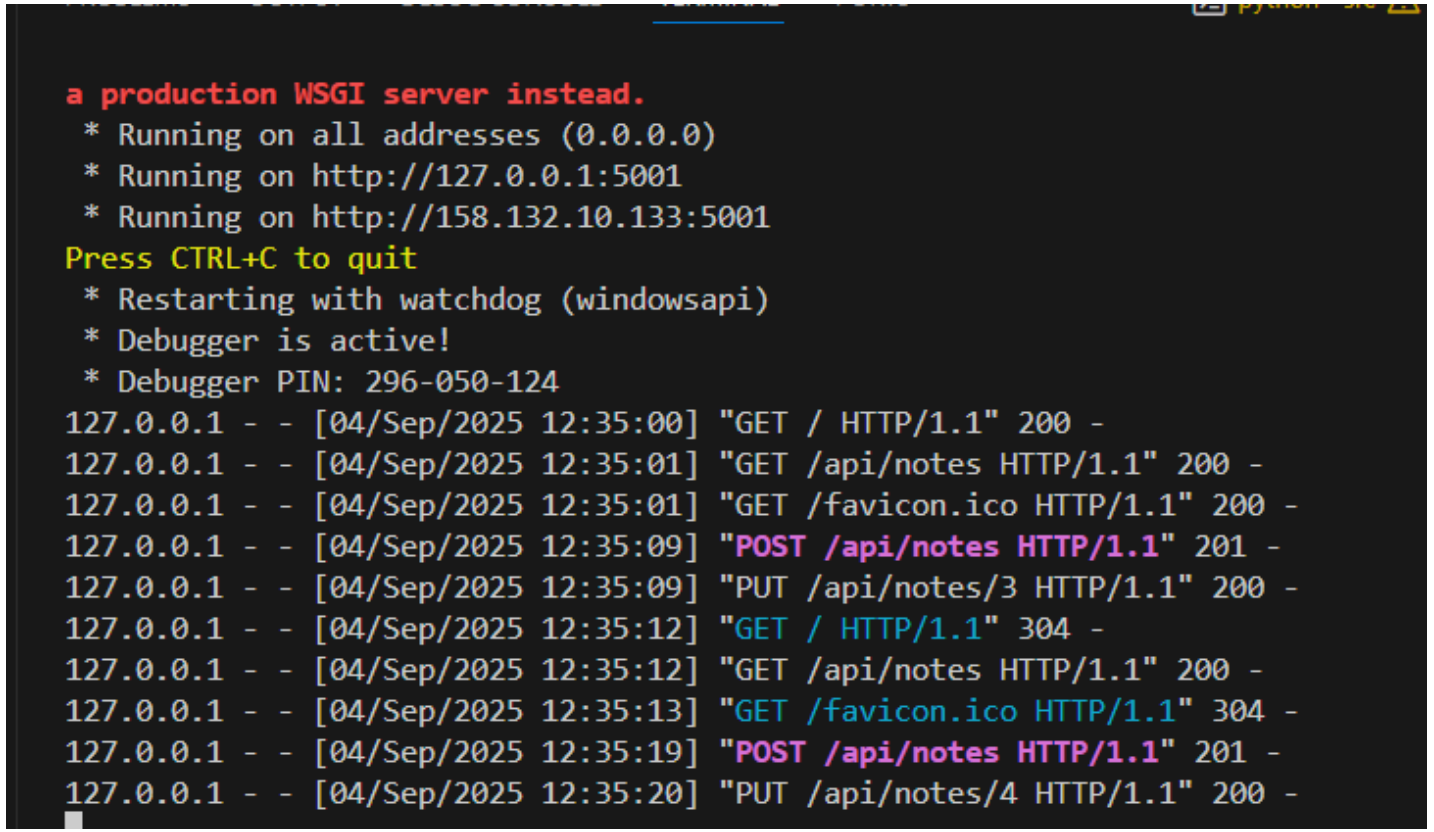
Starting the app

```
PS C:\programs\notetaking-app\src> python main.py
 * Serving Flask app 'main'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use
a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5001
 * Running on http://158.132.10.133:5001
Press CTRL+C to quit
 * Restarting with watchdog (windowsapi)
 * Debugger is active!
 * Debugger PIN: 296-050-124
```

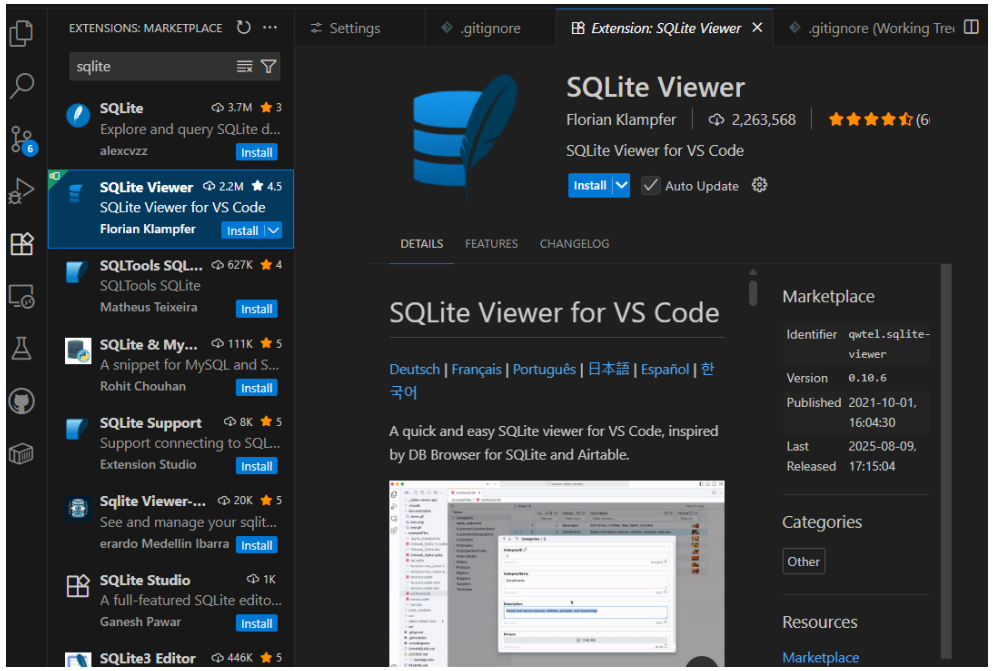
- Test the app using the exposed URL by codespaces

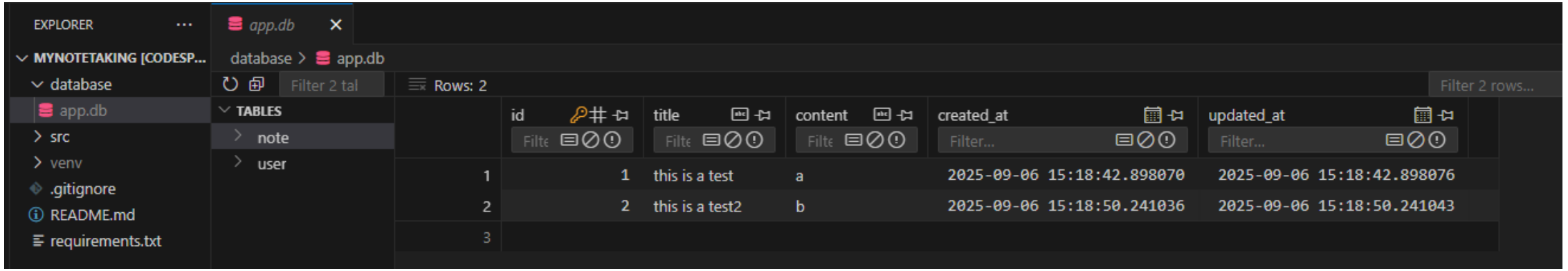


- Test the app and observe the backend logs



- Install “SQLite Viewer” VSCode plugin
- View the database



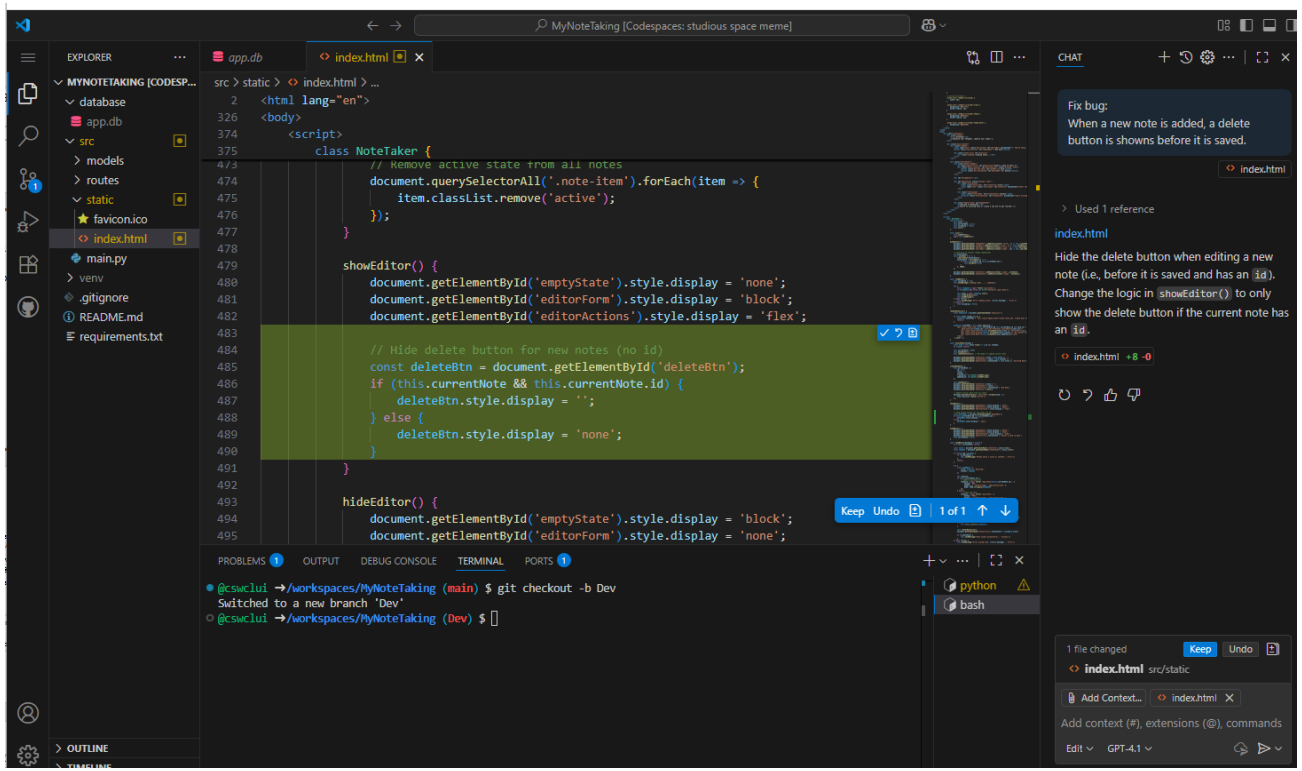


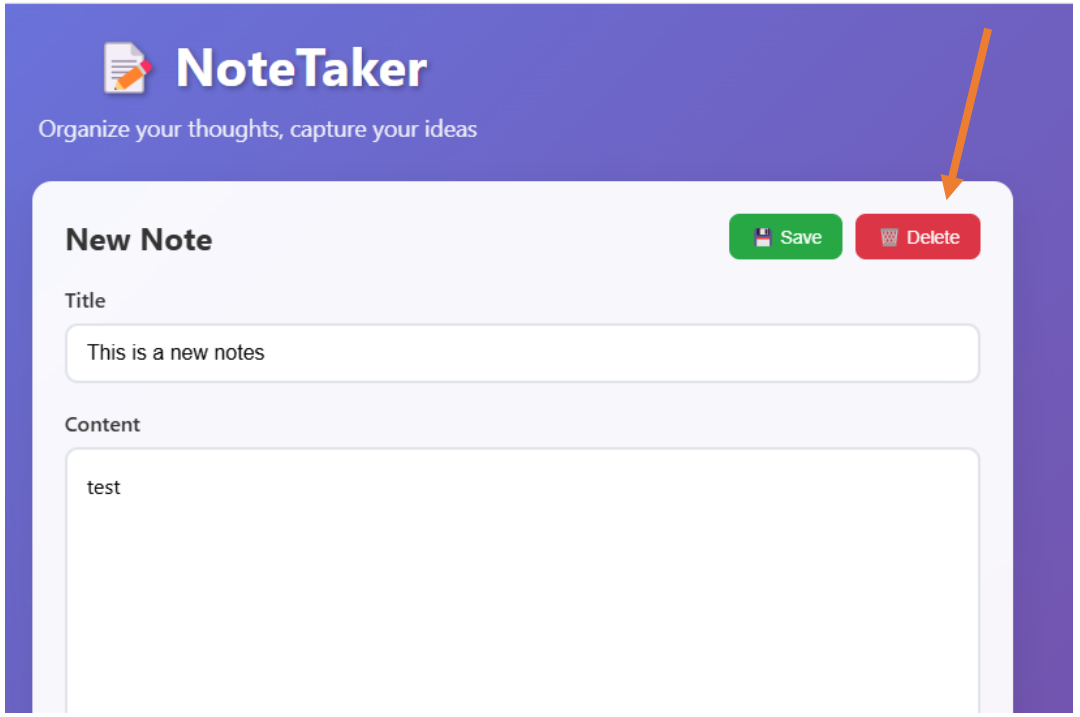
Edit Mode

- You can pick the file(s) you want to change and describe the update in natural language. Copilot will apply inline, review-ready code edits across those files.
- Edit mode is perfect when you know what you want to do but don't necessarily want to write it all out yourself.
- **Exercise**: Use the edit mode to change the background color of the web app
  - select **static/index.html** as context

Exercies: Fix a bug

- Use the Copilot **Edit** mode to fix the bug that delete button should not appear before the notes is saved.





Git

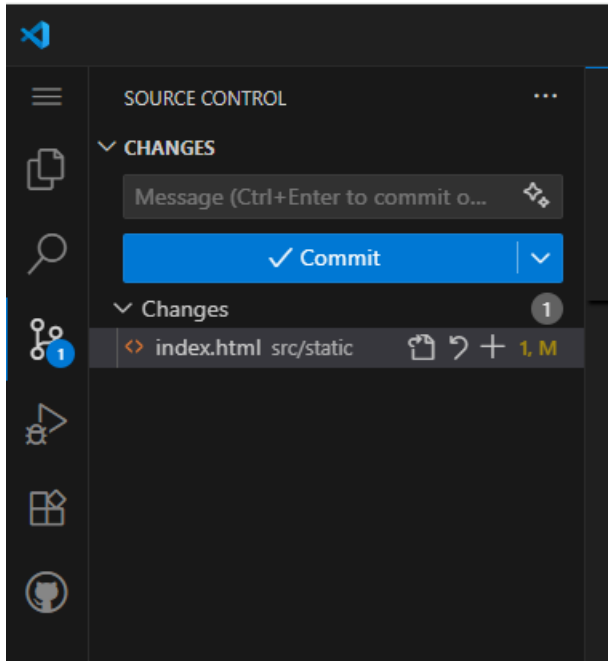
- In GitHub Codespaces, any changes made within the environment will be lost once the environment is deleted because it is temporary.
- To ensure your work is saved, you should first commit your changes to the local Git repository. After committing, push these changes to the remote repository on GitHub.
- This process ensures that all your work is stored safely and can be accessed from any future environment.

- Git: Version control system used to track changes in source code during software development
- Commit: A commit in Git is a snapshot of your changes in the repository.

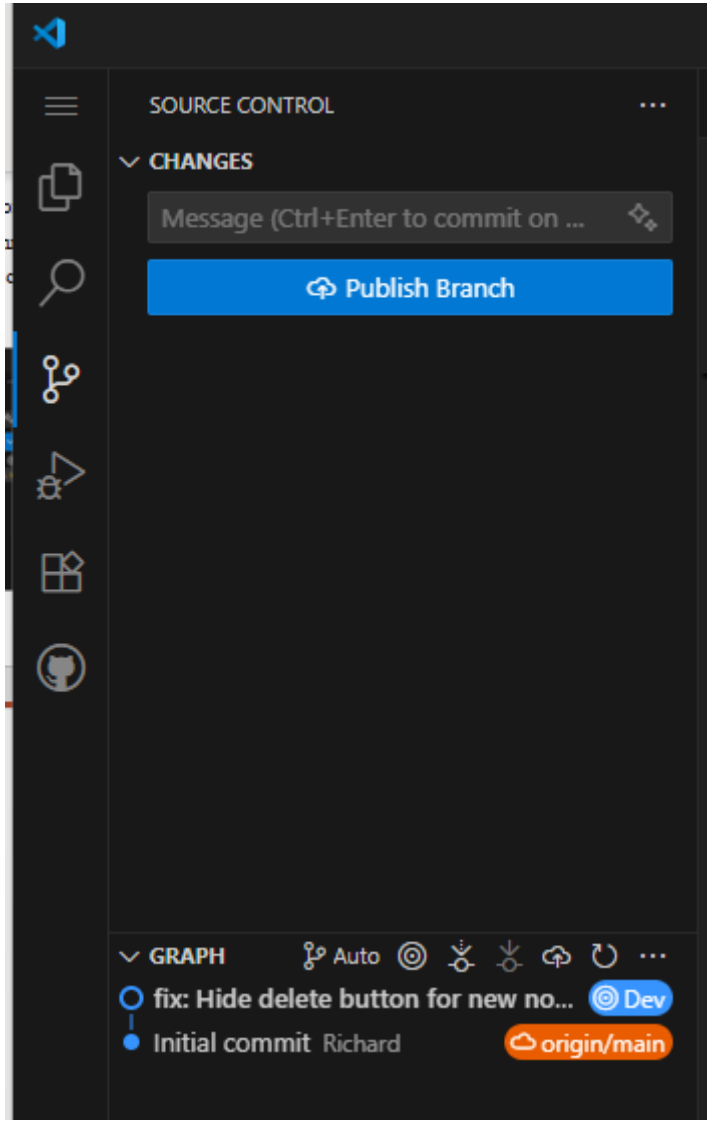
Git Branch

- **Branch**: A parallel version of the repository. It allows you to work on different features, fixes, or experiments independently from the main codebase.
- In Git, the use of branches allows you to diverge from the main line of development and continue to do work without affecting the main branch.
  - E.g., you may work on new features, bug fixes, or experiments without affecting the main codebase. Multiple developers can work on different tasks simultaneously without interfering with each other

- Create a new branch “Dev” and checkout to the branch
- Stage and commit the change with a descriptive commit message
- Push the new commit to the GitHub repository

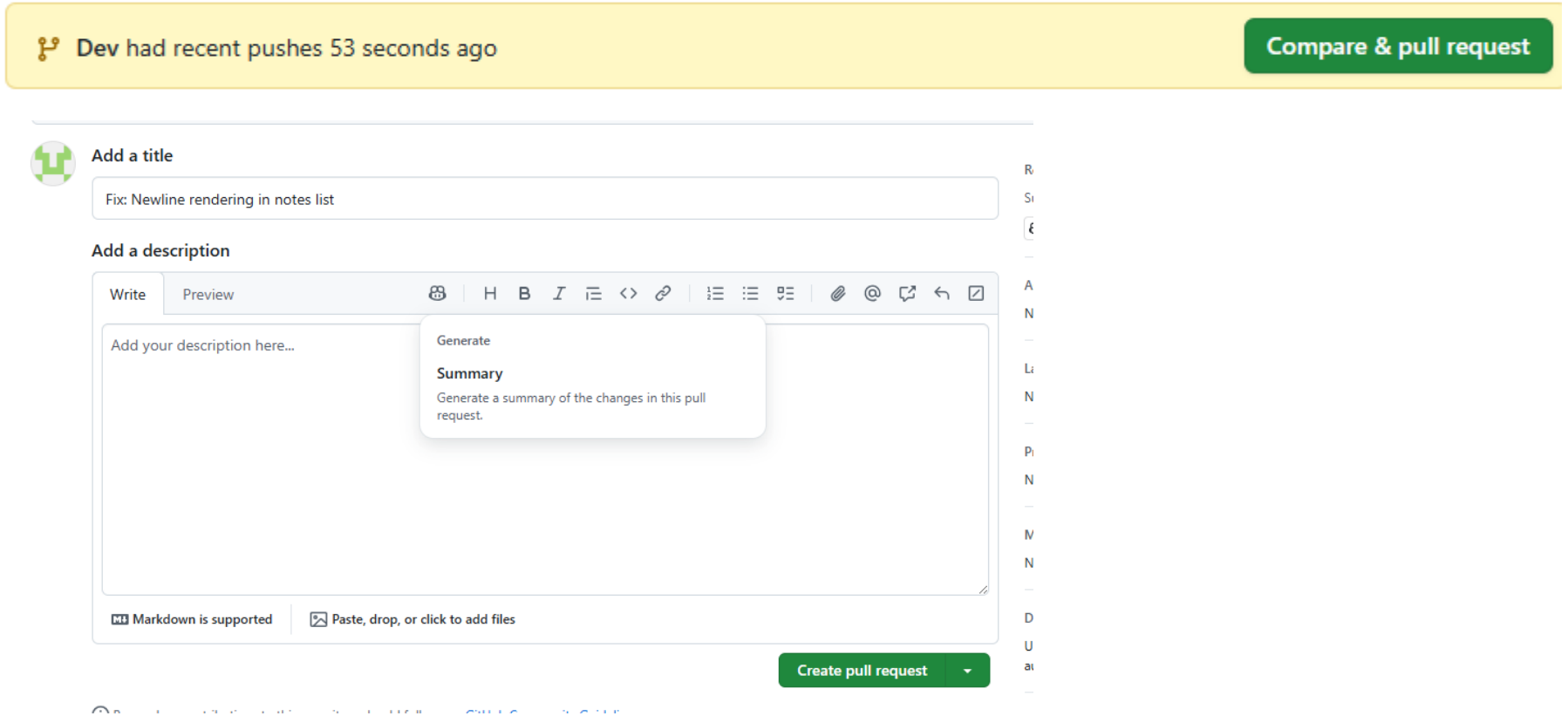



```
# Create and checkout to a new branch named "Dev"
git checkout -b Dev

# Stage your changes
git add .

# Commit with a descriptive message
git commit -m "fix: Hide delete button for new notes in the editor"

# Push the new branch and commit to GitHub
git push origin Dev
```

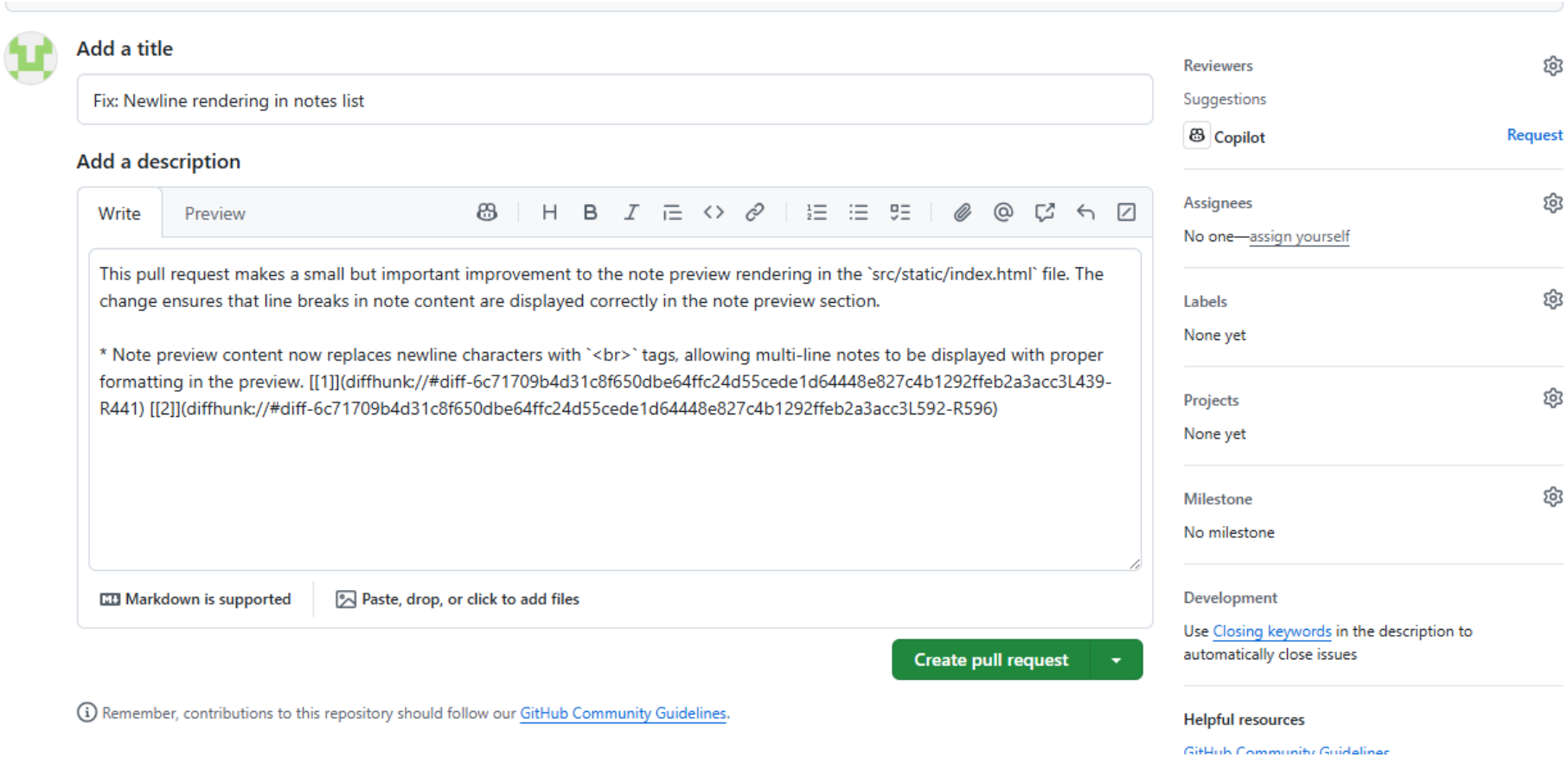



```
# Push the new branch and commit to GitHub
git push origin Dev
```

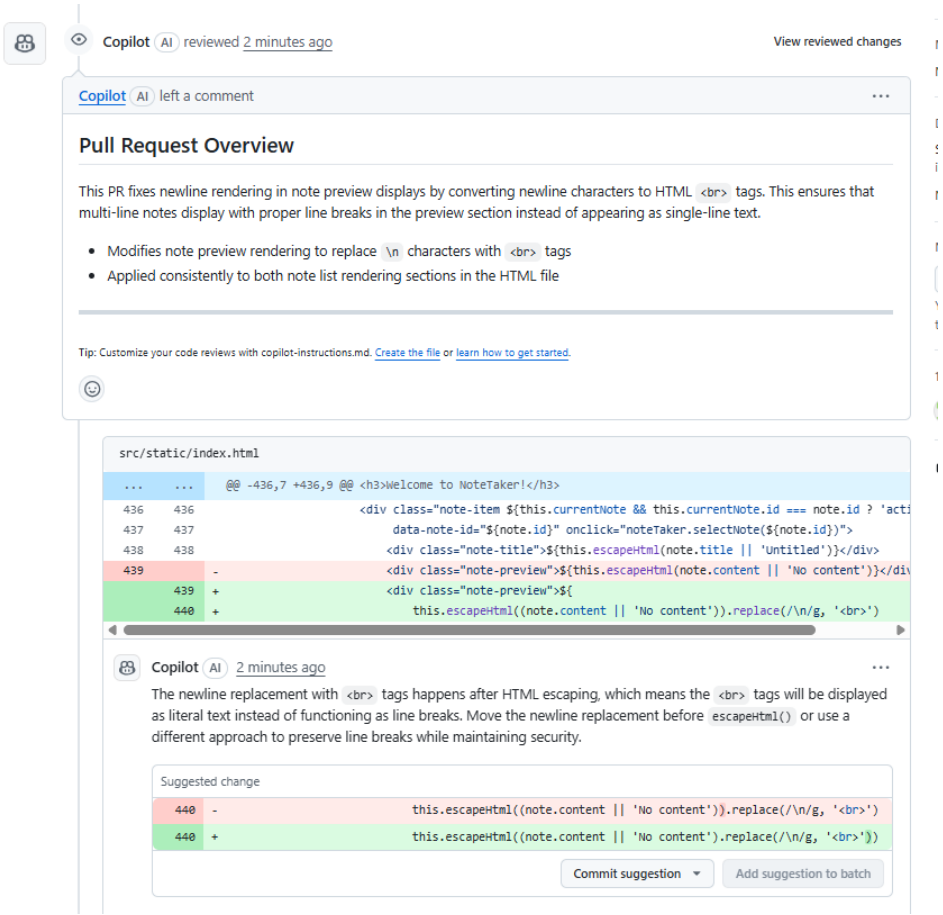
- Now, generate a pull request
  - A pull request is a formal proposal to merge code changes from one branch into another, allowing for review, discussion, and approval before integration.
- Request Copilot to generate a summary

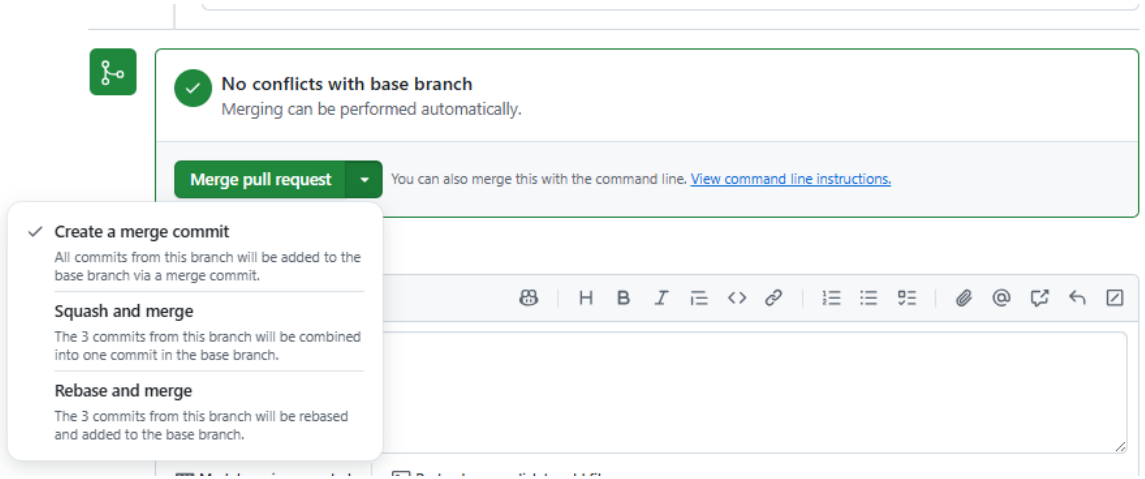


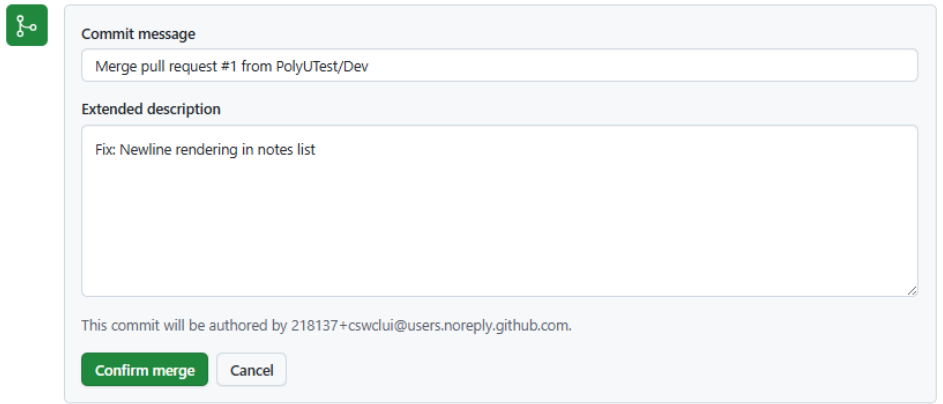
- You may request a team member/copilot to review the code change



- Review the suggestion of the Copilot coding agent.
- Merge the pull request

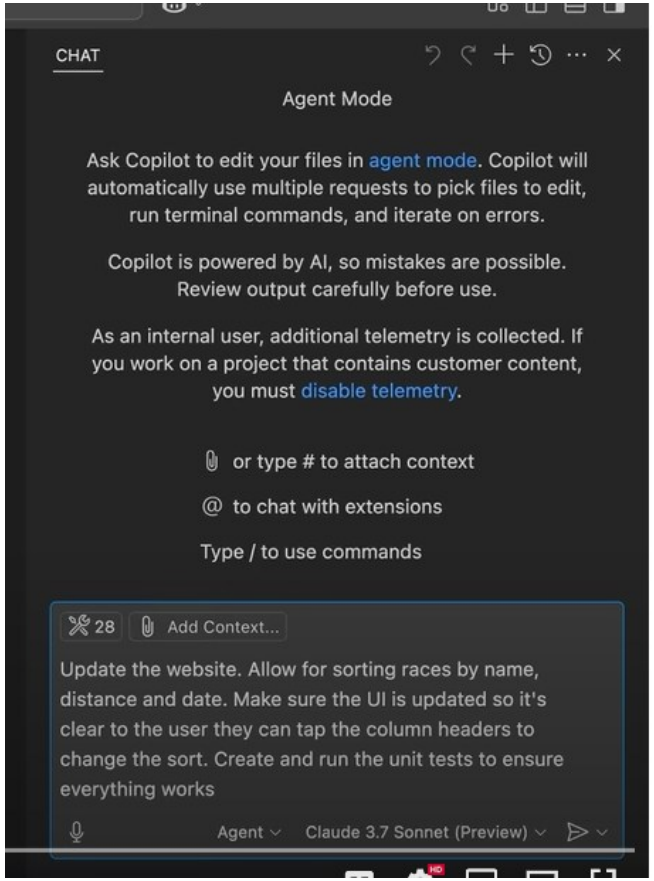






Agent mode

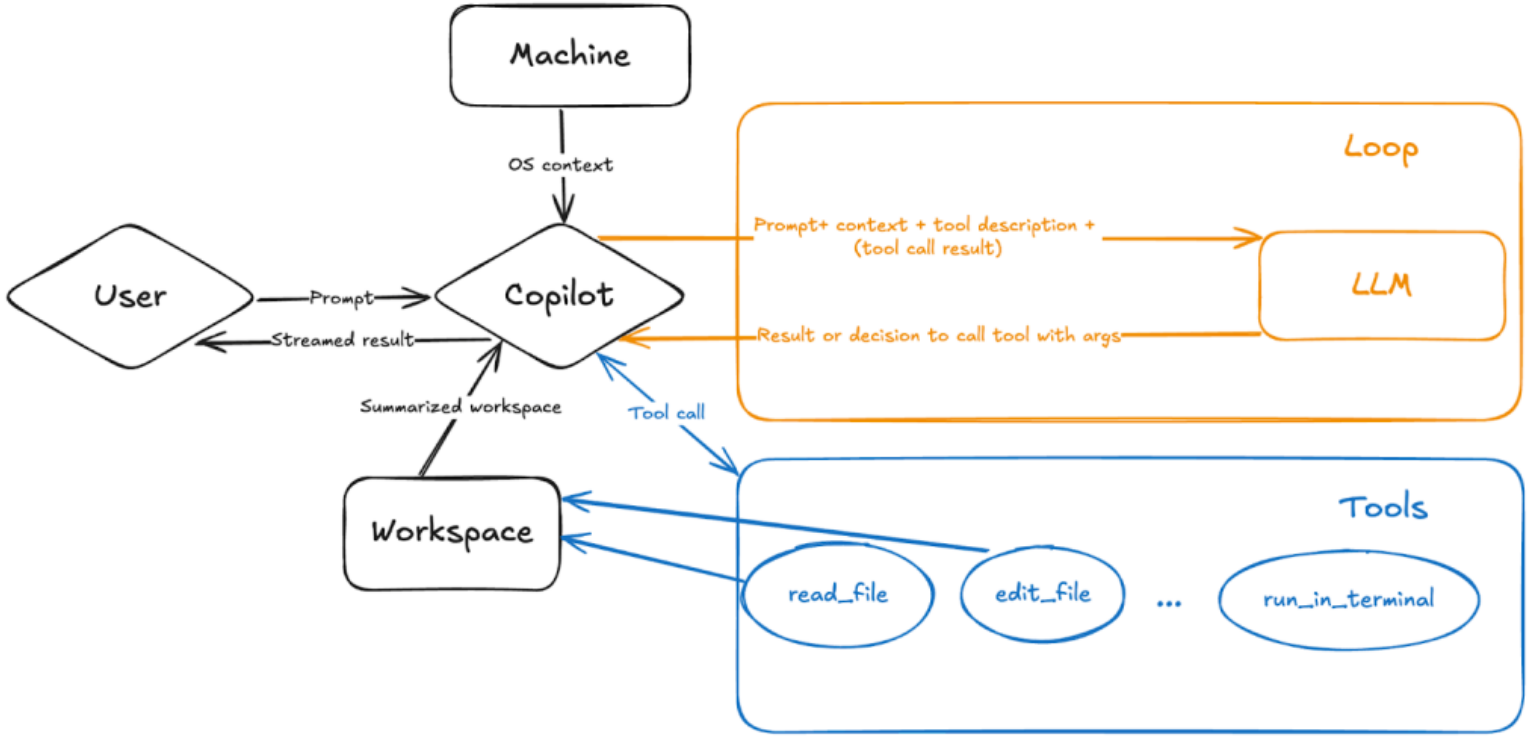
- Use natural language to specify a high-level task
- Let AI it autonomously plans and executes a multi-step workflow to complete those tasks across your codebase
  - Analyze your codebase to grasp the full context.
  - Plan and execute multi-step solutions.
  - Run commands or tests.
  - Reach out to external tools for specialized tasks (e.g. MCP)
  - Monitoring the results of code changes, build outcomes, test failures, and terminal outputs.
  - Iterating and refining the changes autonomously until the task is successfully completed or until it requires input from you.



Video

https://github.blog/ai-and-ml/github-copilot/agent-mode-101-all-about-github-copilots-powerful-mode/

How does agent mode in GitHub Copilot work?

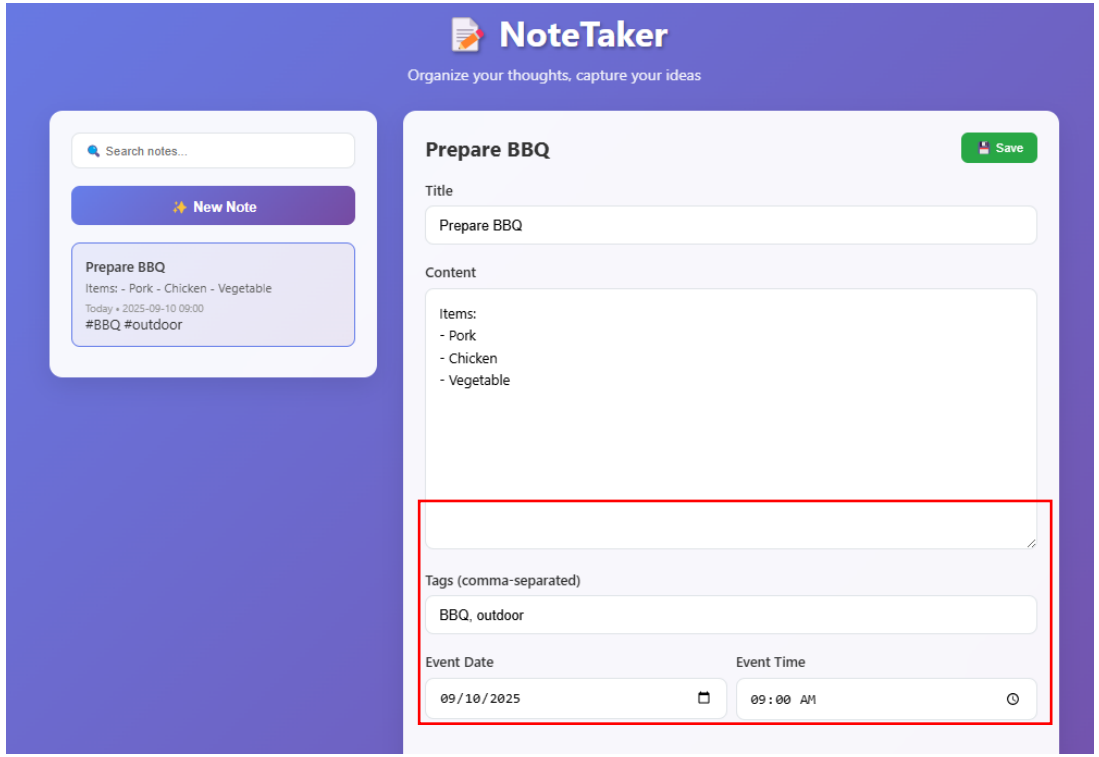


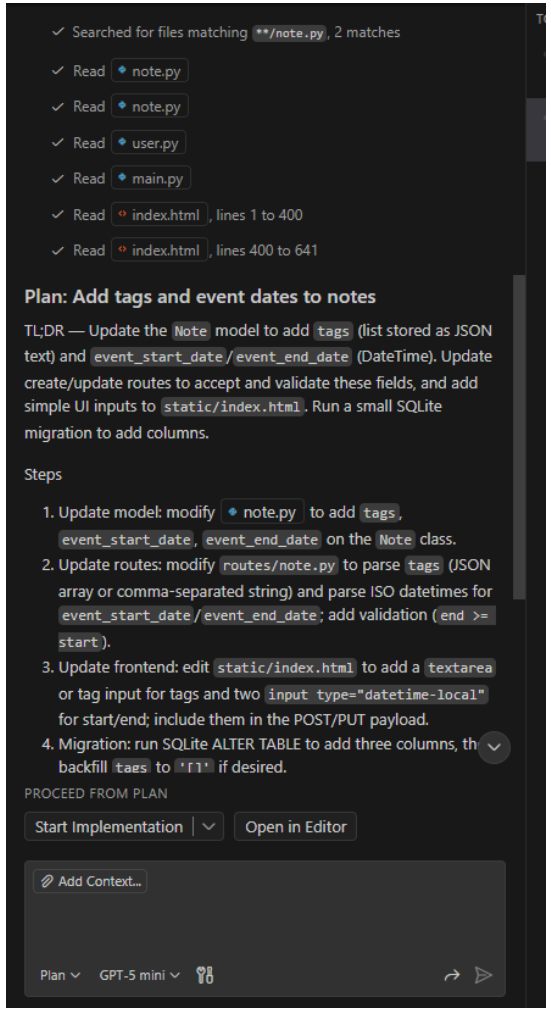
- User query is augmented with workspace structure summary, tool descriptions, other contexts.
- Copilot parses the question & plans resolution
- AI begins working and uses tools autonomously to reach the goal
  - Searching workspace, Reading/Editing files, Running terminal commands, Handling errors and applying fixes
- When errors or issues (like syntax errors or failed tests) are encountered, AI will analyzes the problem and makes changes to fix those errors and move closer to the desired result.
  - E.g. editing code, running new commands, or using different tools

Add new fields for the notes item

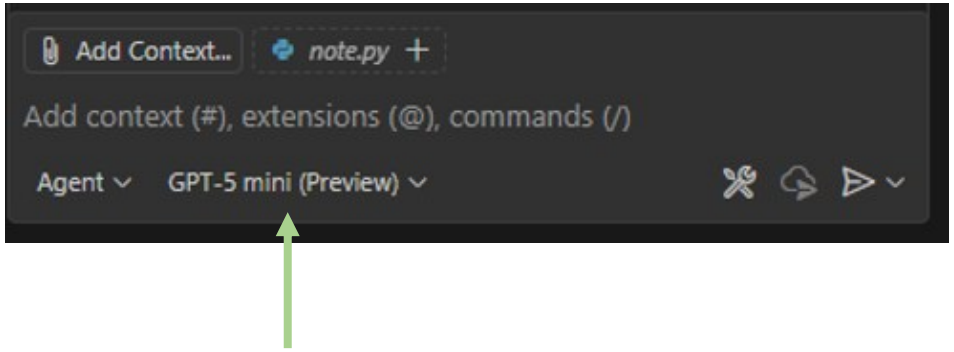
- First, use the "Plan" mode to generate the plan

Task: add the following fields in the notes



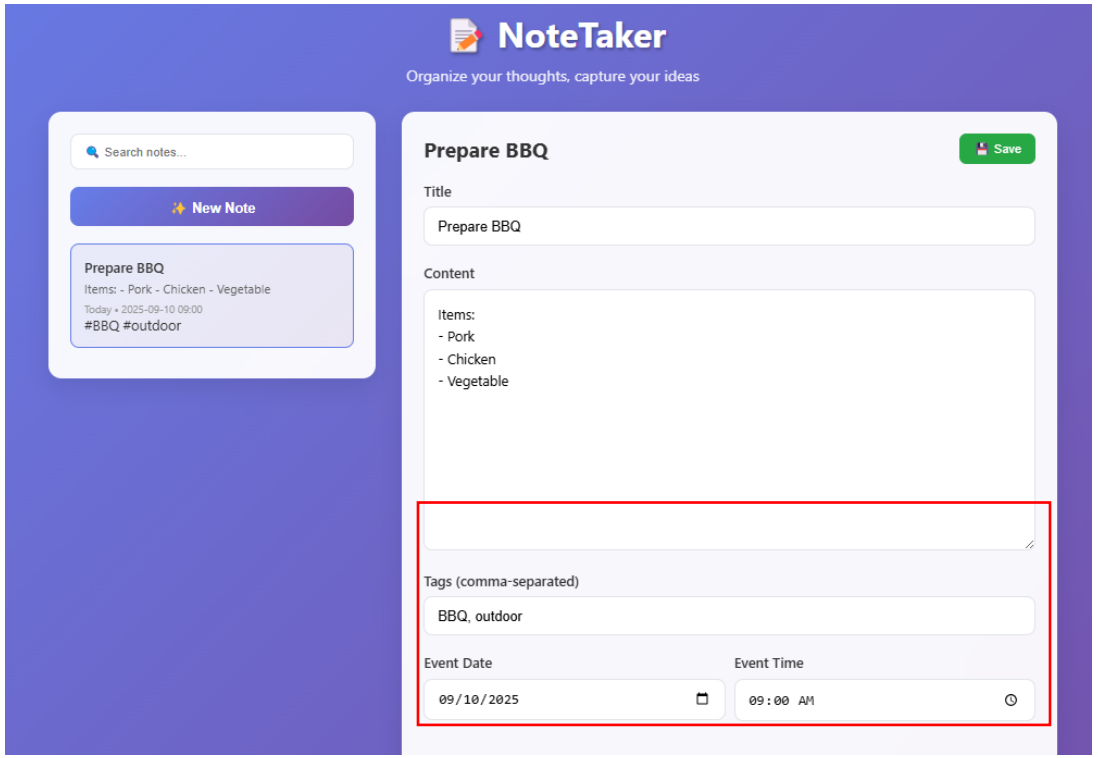


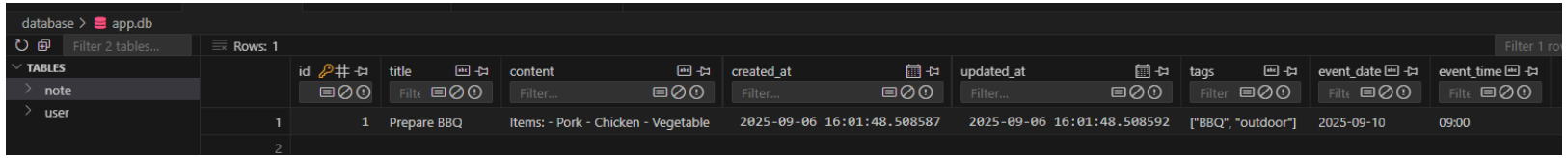
- Use GitHub Copilot "**Agent**" mode to implement it
- Observe how the agentic coding works.



Use GPT5-mini or Claude 4.5.

Task: add the following fields in the notes





Agent mode vs edit mode

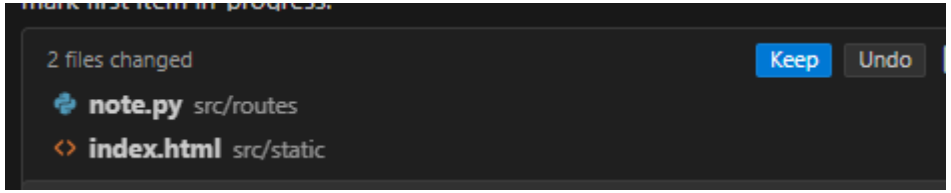
- **Edit scope**: agent mode autonomously determines the relevant context and files to edit. In edit mode, you need to specify the context yourself.
- **Task complexity:** agent mode is better suited for complex tasks that require not only code edits but also the invocation of tools and terminal commands.
- **Duration:** agent mode involves multiple steps to process a request, so it might take longer to get a response.
  - E.g., to determine the relevant context and files to edit, determine the plan of action.
- **Self-healing:** agent mode evaluates the outcome of the generated edits and might iterate multiple times to resolve intermediate issues.
- **Request quota:** in agent mode, depending on the complexity of the task, one prompt might result in many requests to the backend

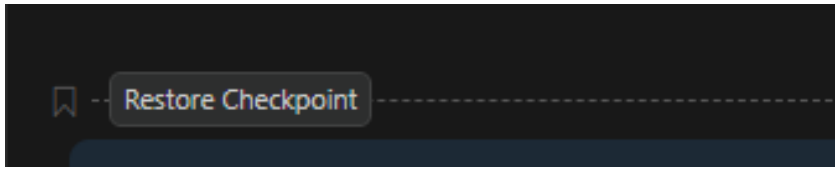
Which mode to use?

- Agent mode might
  - Touch a file you thought you had agreed to leave alone.
  - Suggest running a command you don't expect.
  - It takes longer for reasoning and performing the different steps, especially on bigger projects.
- Sometimes you'll be working on part of a codebase that needs to be handled with more precision.
  - E.g. Treaking a couple of files or making a targeted change in a sensitive system
  - Ask or edit mode might be the better fit

Undoing the changes generated by AI

- “Undo” the AI changes
- “Restore Checkpoint” in the GitHub Copilot Chat
- Using “git reset” Command
  - Soft, Mixed vs. Hard reset



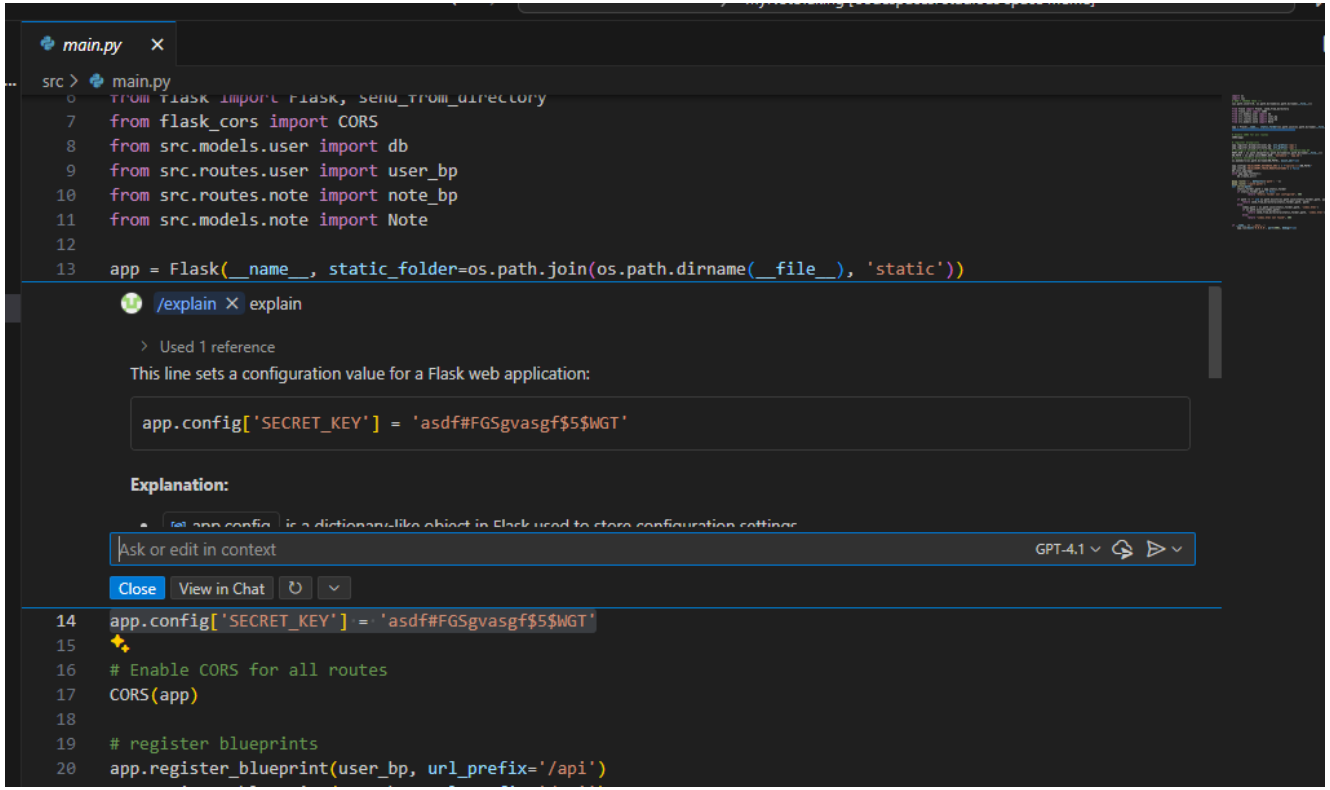


Exercise

- Fix the remaining bugs of the app.
- Add new features to the app.
- Remark:
  - After the AI makes the code changes, you should test to see if the code changes is working correctly
  - Commit each change in the “Dev” branch
  - Push the commits and update the remote repository (GitHub)

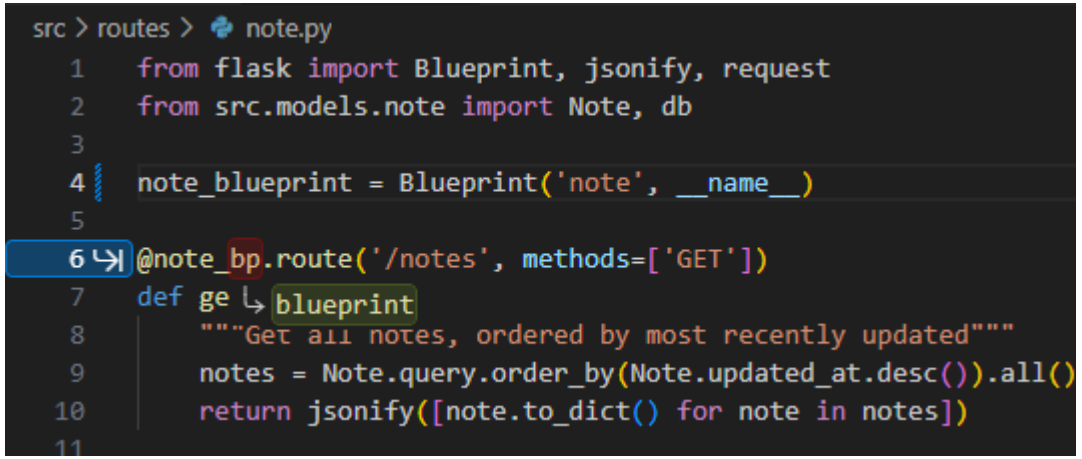
Inline chat

- Ask questions and get suggestions directly in the editor or get help with shell commands within the integrated terminal.
- Open editor inline chat by using the **Ctrl+I** keyboard shortcut



Next edit suggestions (NES)

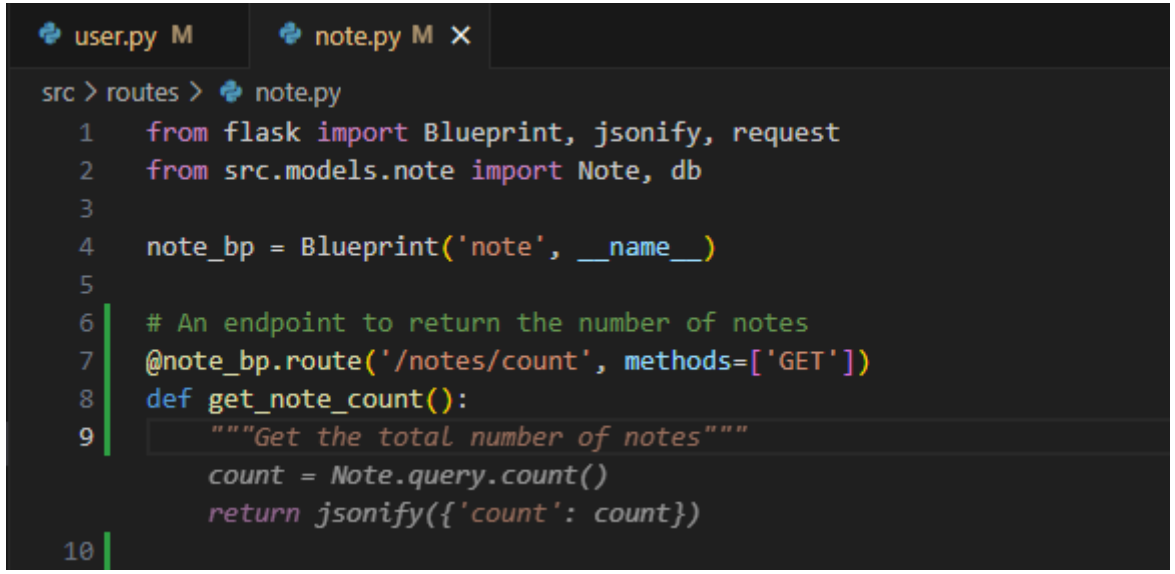
- Based on the edits you're making, NES both predicts the location of the next edit you'll want to make and what that edit should be.
- Press tab to accept the suggestion



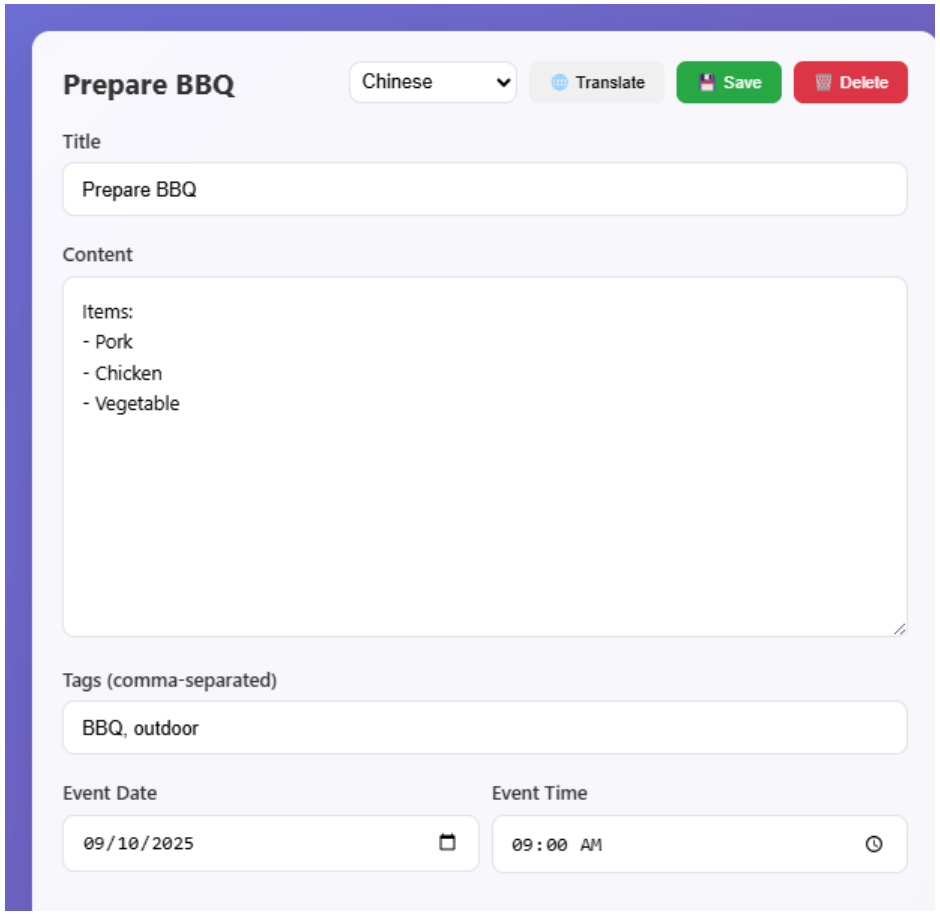
https://code.visualstudio.com/docs/copilot/ai-powered-suggestions

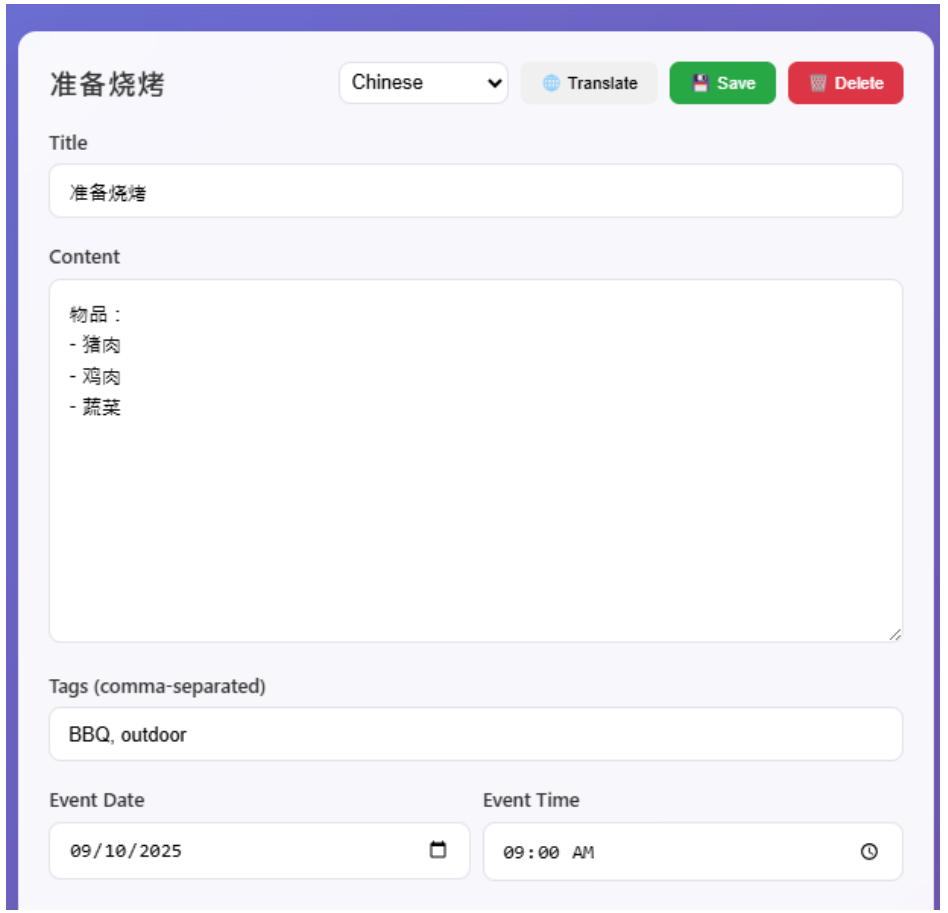
Code completions

- Start typing in the editor, and Copilot provides code suggestions that match your coding style and take your existing code into account.
- Generate suggestions from code comments
- Press **tab** to accept the suggestion



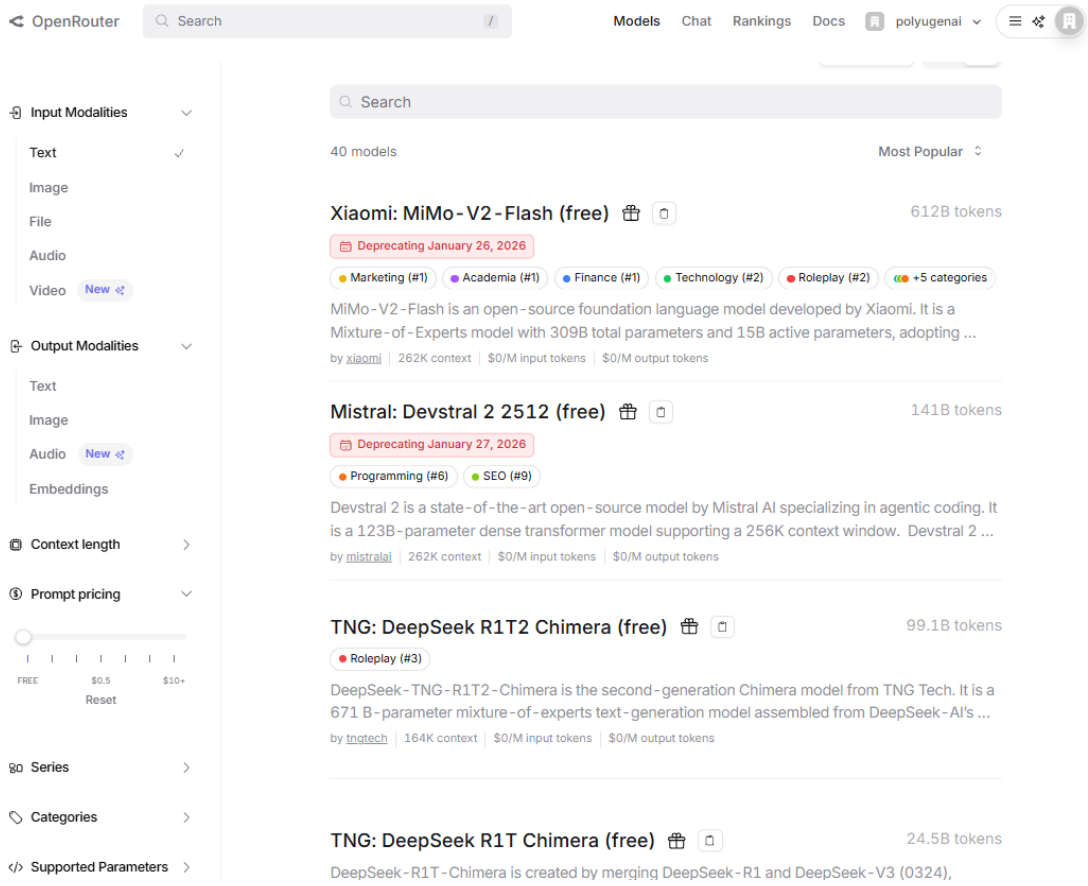
Let's develop a new feature to translate the notes using GPT 4.1-mini





Openrouter APIs

- Free models are available
- Free quote
  - Budget can get to -$1



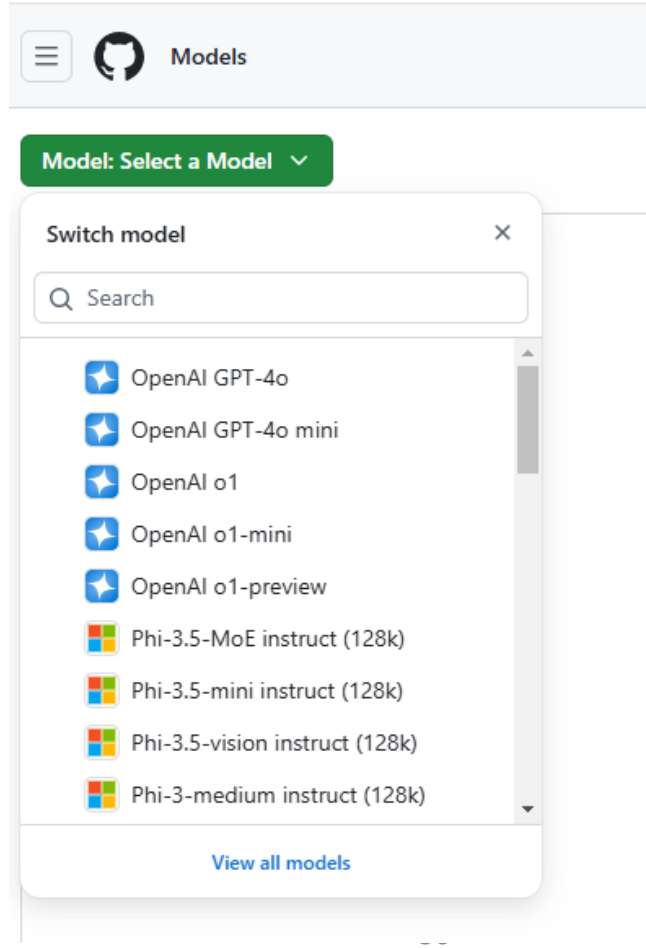
https://openrouter.ai/models

GitHub Models

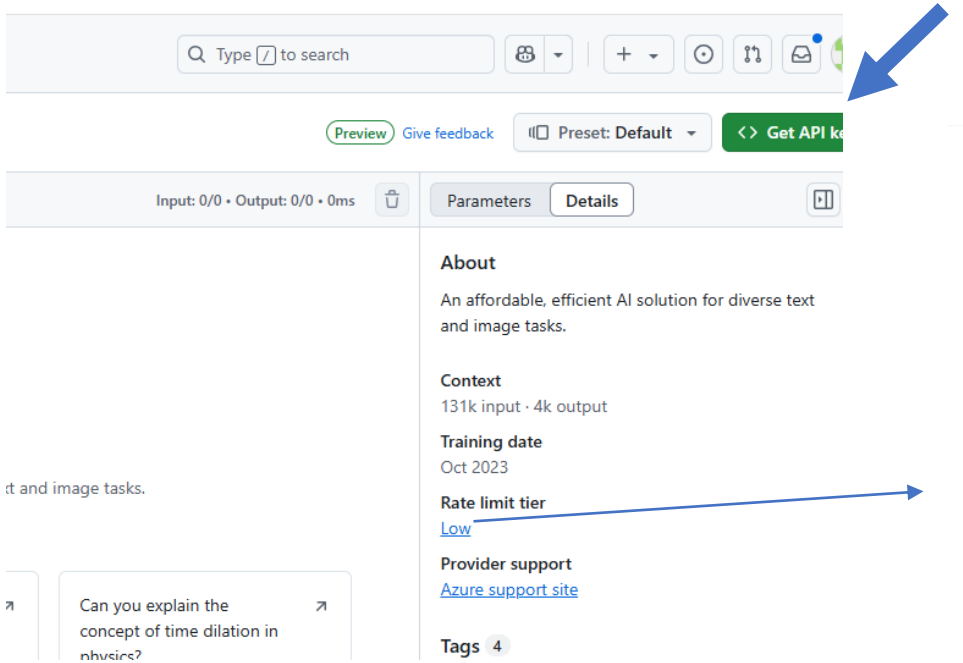
- You can use GitHub Models to find and experiment with AI models for free.

1.Go to github.com/marketplace/models.

2.Click **Model: Select a Model** at the top left of the page.



- Visit https://github.com/marketplace/models/azure-openai/gpt-4-1-mini/playground



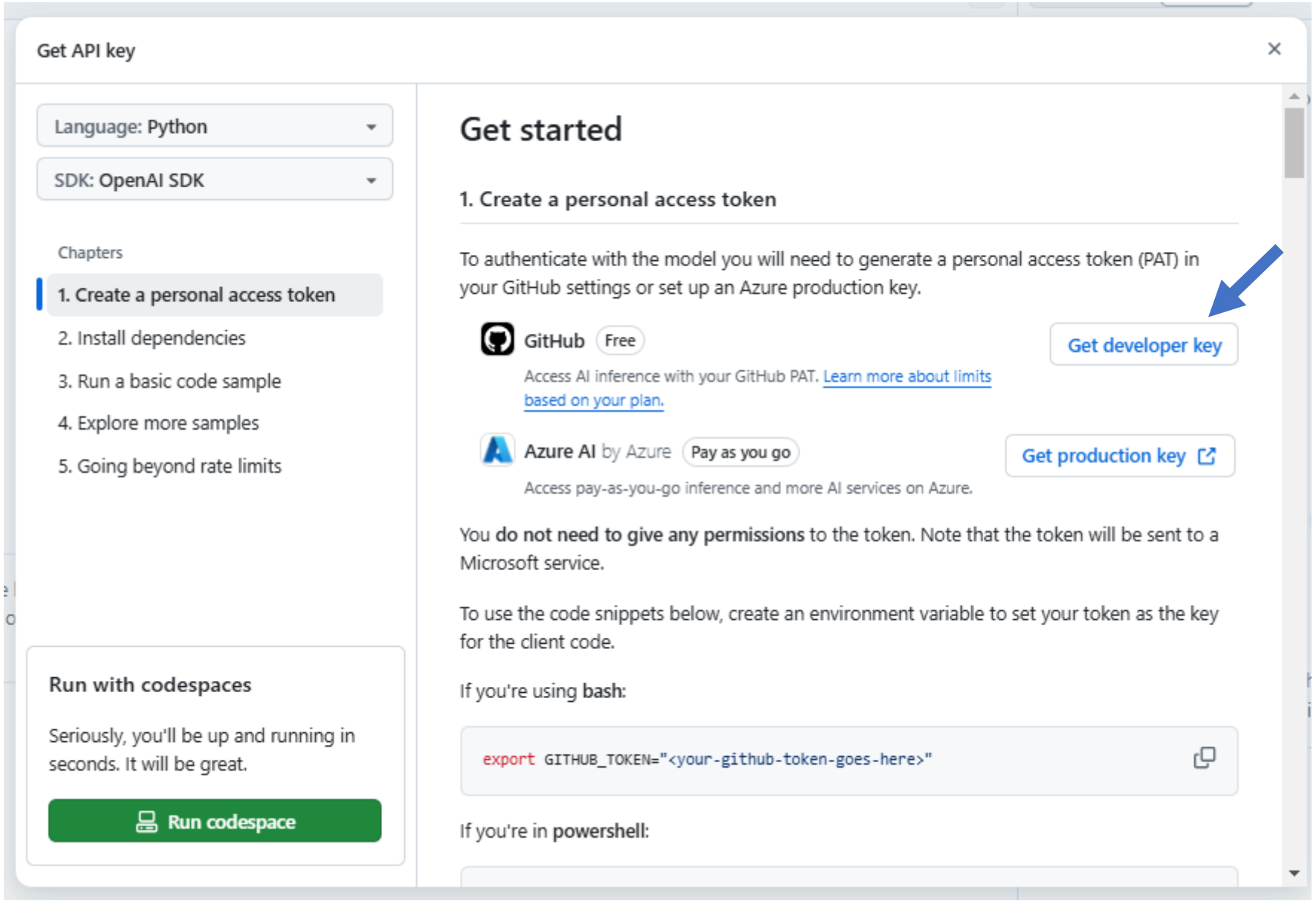
Rate limits

The playground and free API usage are rate limited by requests per minute, requests per day, tokens per request, and concurrent requests. If you get rate limited, you will need to wait for the rate limit that you hit to reset before you can make more requests.

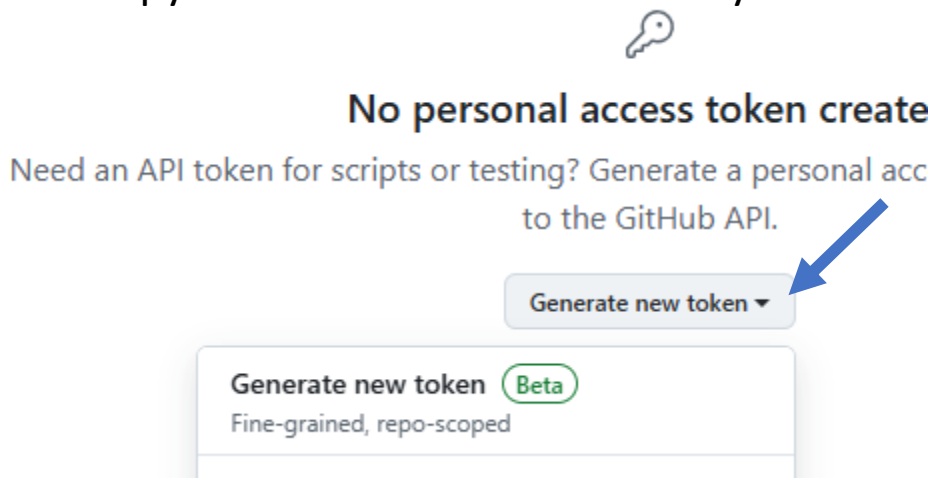
Low, high, and embedding models have different rate limits. To see which type of model you are using, refer to the model's information in GitHub Marketplace.

Rate limit tier	Rate limits	Free and Copilot Individual	Copilot Business	Copilot Enterprise
Low	Requests per minute	15	15	20
	Requests per day	150	300	450
	Tokens per request	8000 in, 4000 out	8000 in, 4000 out	8000 in, 8000 out
	Concurrent requests	5	5	8
High	Requests per minute	10	10	15
	Requests per day	50	100	150
	Tokens per request	8000 in, 4000 out	8000 in, 4000 out	16000 in, 8000 out
	Concurrent requests	2	2	4

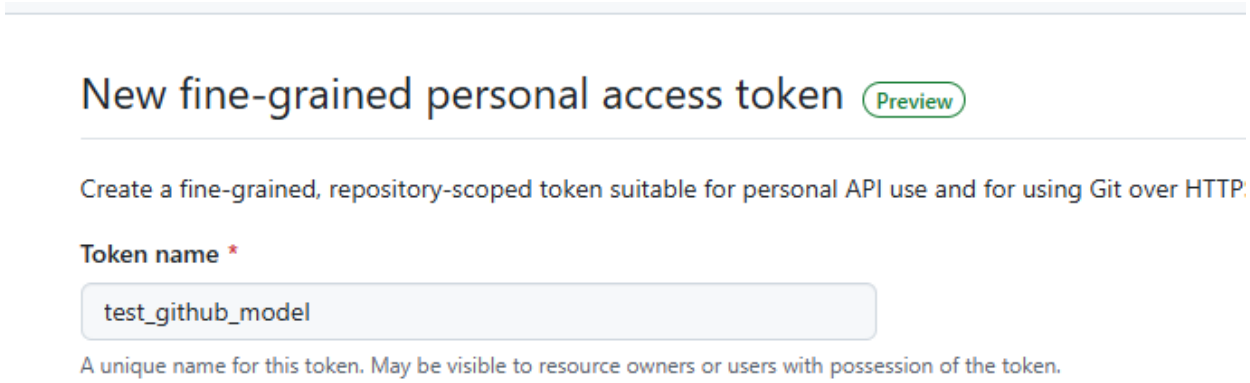
- Follow the steps to get an API Key



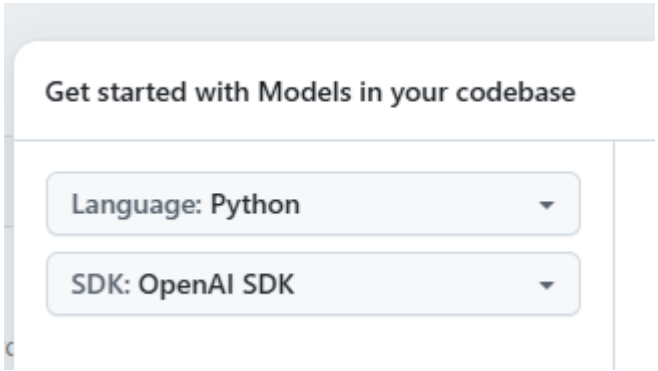
Click “Generate Token”.
Copy the token and store securely.



- For security reasons, the API key should never be hardcoded in source code because it risks accidental exposure if the code is shared, published, or pushed to a public repository To secure the API key,
- We may store the API key in a .env file or environment variable
- Ensure the .env file is listed in .gitignore so it is not committed to version control.
- To load it from the .env file or environment variable in your code using a package like python-dotenv, then access it with os.getenv("GITHUB_TOKEN").



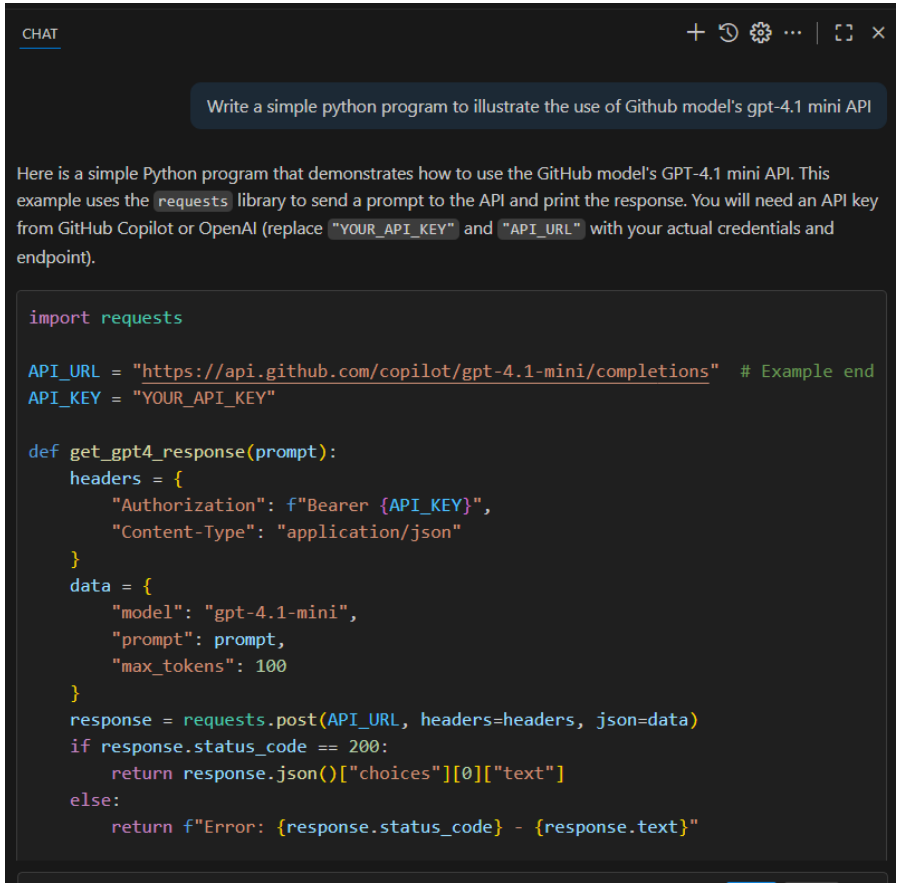
Choose Python as language and OpenAI SDK.
Follow the remaining steps to configure the environment and run the sample program with the API key




```python
import os
from openai import OpenAI

token = os.environ["GITHUB_TOKEN"]
endpoint = "https://models.github.ai/inference"
model = "openai/gpt-4.1-mini"

client = OpenAI(
    base_url=endpoint,
    api_key=token,
)

response = client.chat.completions.create(
    messages=[
        {
            "role": "system",
            "content": "You are a helpful assistant.",
        },
        {
            "role": "user",
            "content": "What is the capital of France?",
        }
    ],
    temperature=1.0,
    top_p=1.0,
    model=model
)

print(response.choices[0].message.content)
```

Limitations: Using LLM for coding

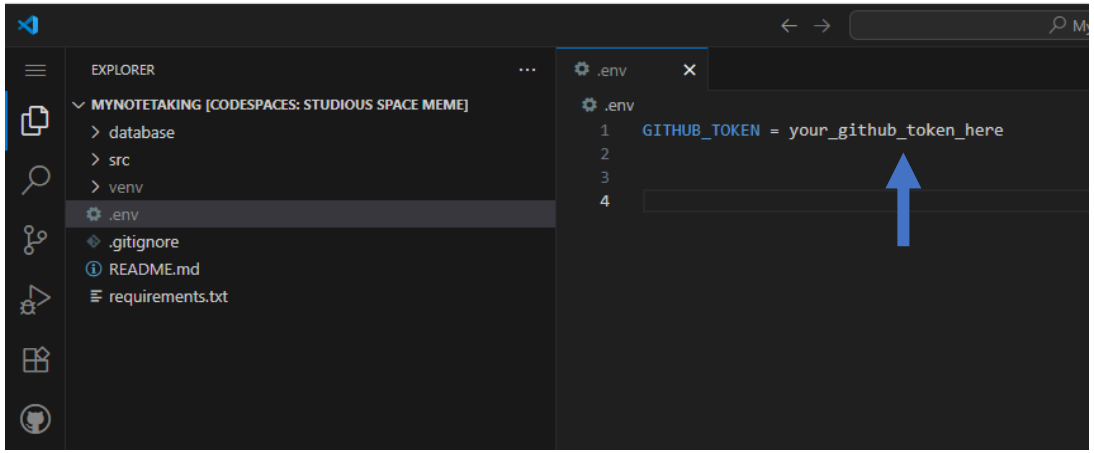
- Large language models (LLMs) are trained on large code corpora collected at a fixed point in time, LLMs often lack real-time updates on library evolution and API changes.
- As a result, they may suggest code that use deprecated libraries or outdated APIs that may no longer function correctly with the latest versions of libraries or frameworks.
- The generated code is more prone to hallucination, and might fail to compile or run, resulting in compatibility issues, bugs, or even security vulnerabilities.

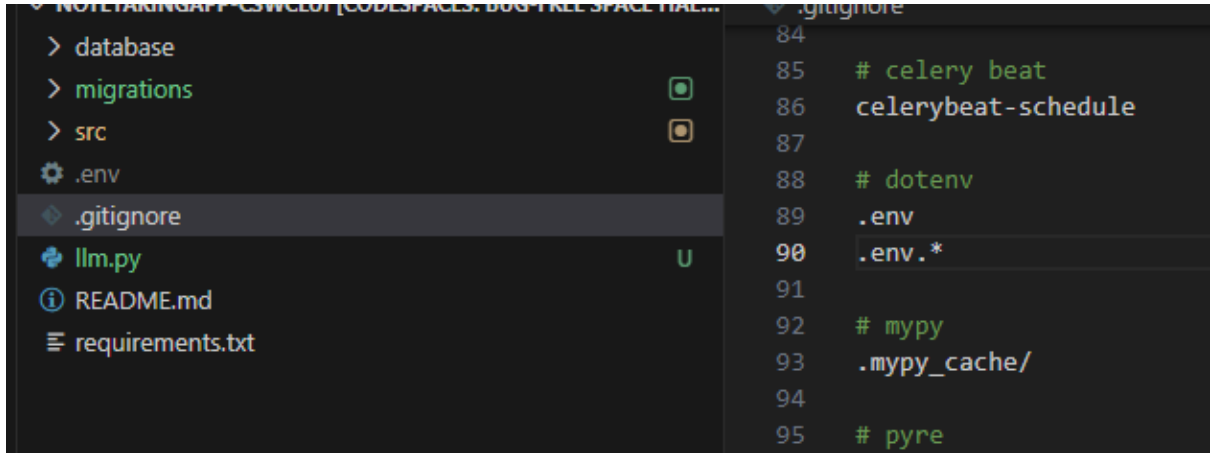


Is the code correct?

Storing API keys/credentials

- Store the GitHub Model Credential in .env file
- Make sure the **.env** is specified **in .gitignore** file and the **.env** is not tracked by git.





Define the following program

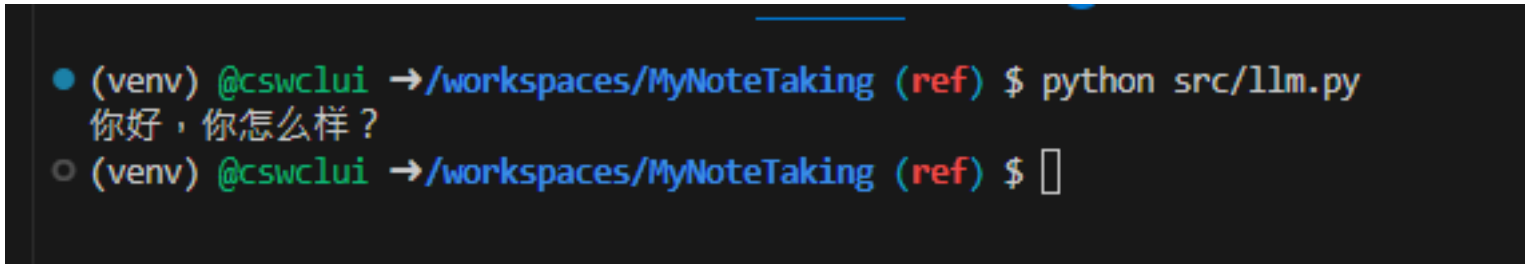
src/llm.py

Instead of asking LLM to complete the feature on its own, we write our code to guide LLM in the code generation.

```
# import libraries
[Add your code here]

load_dotenv()  # Loads environment variables from .env
token = os.environ["GITHUB_TOKEN"]
endpoint = "https://models.github.ai/inference"
model = "openai/gpt-4.1-mini"

# A function to call an LLM model and return the response
def call_llm_model(model, messages, temperature=1.0, top_p=1.0):
    client = OpenAI(base_url=endpoint,api_key=token)
    response = client.chat.completions.create(
        messages=messages,
        temperature=temperature, top_p=top_p, model=model)
    return response.choices[0].message.content

# A function to translate to target language
[Add your code here]


# Run the main function if this script is executed
[Add your code here]
```

Run the program

- Add the library in requirements.txt
  - openai==1.106.1
  - dotenv==0.9.9
- Install the library in the virtual environment
  - pip install -r requirements.txt
- Run the app



Implement the feature

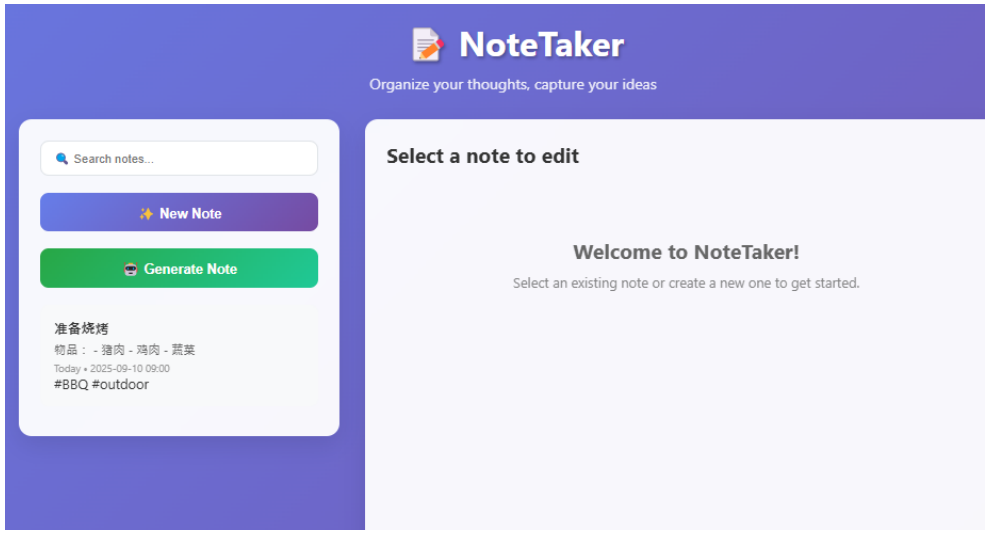
- Add src/llm.py to Copilot chat and use it for context
- Use the copilots “Agent mode” to reference the code and use the defined function to implement the translation feature by modifying the frontend and backend API.

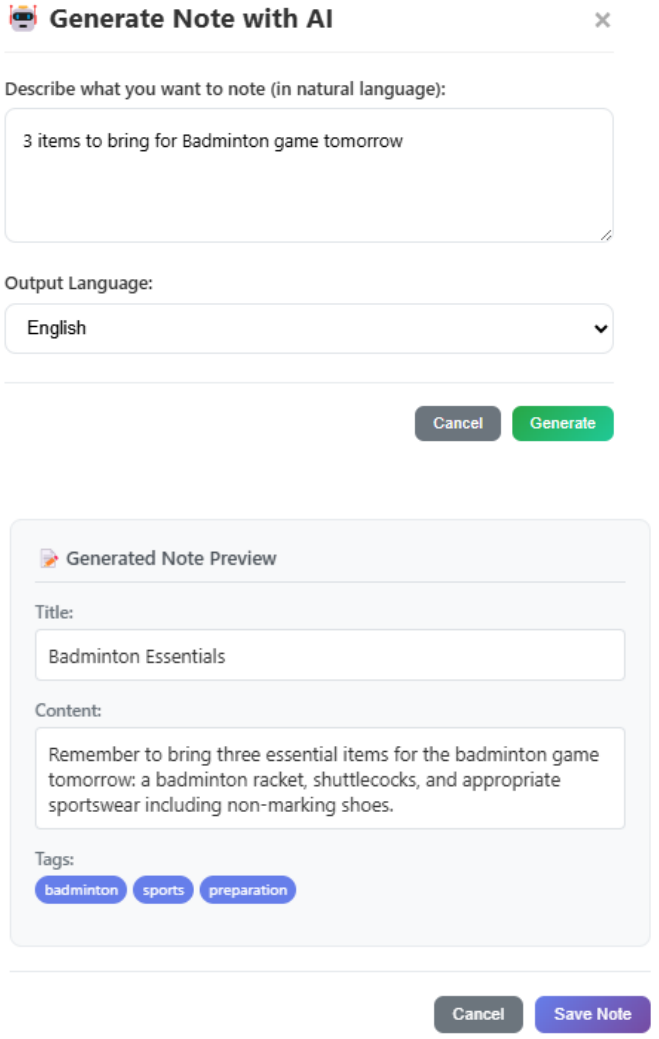
Commit change in Dev branch

- Use the GitHub Copilot Agent mode to add the translation function to the App
- Stage the changes
- Commit with message **"feat: Translate notes"**

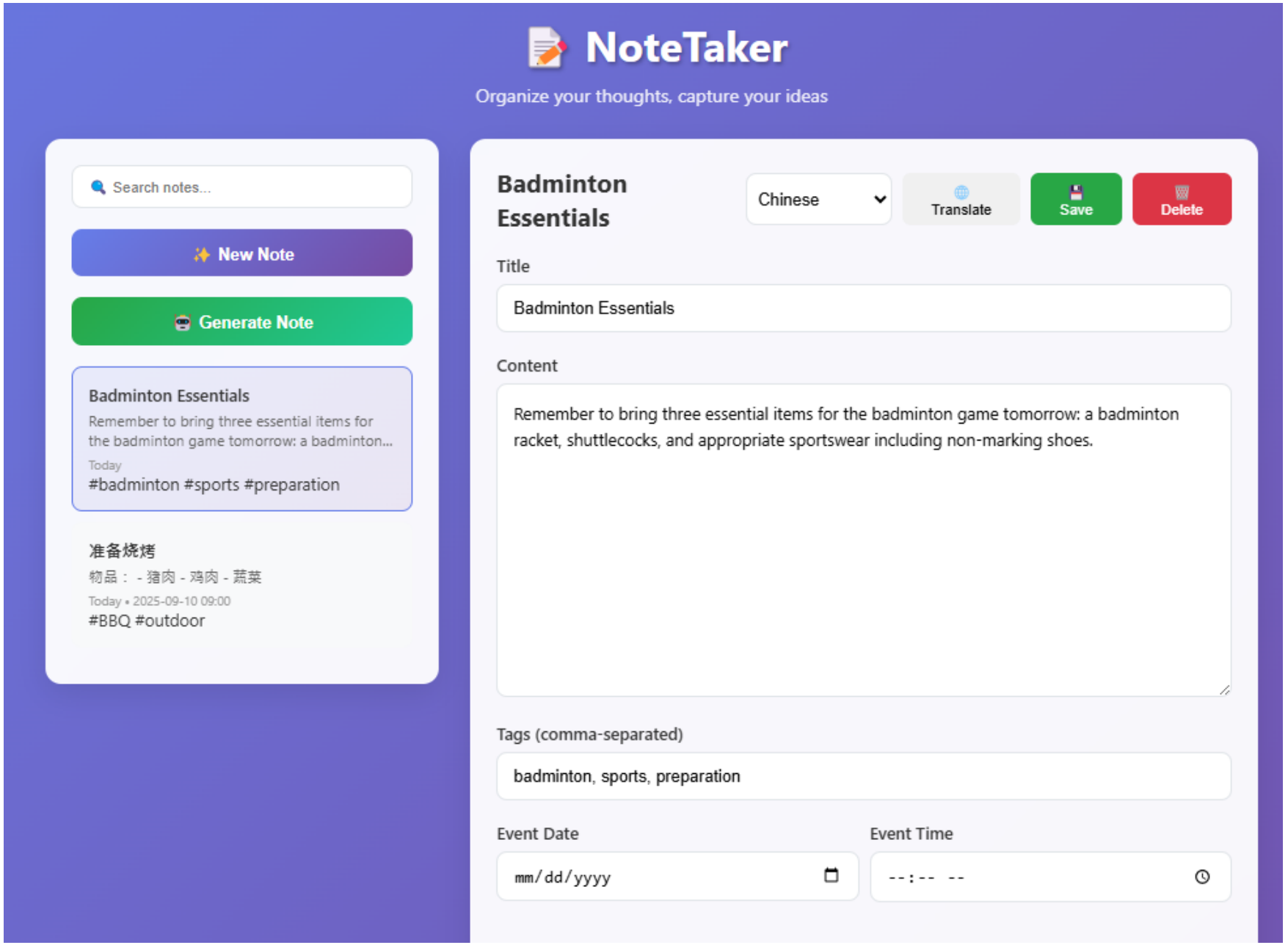
```
# Create and checkout to a new branch named "Dev“
# If you already have the branch created before, use “git checkout Dev” to change to the branch

git checkout -b Dev

# Stage your changes and commit
git add .
git commit -m "feat: Translate notes"

# Push the new branch and commit to GitHub
git push origin Dev
```

Add feature: “Generate Notes”





Allow the user to input the notes in natural language. LLM is used to generate the title, contents of the notes with three suggested tags



The AI generated notes with title, content, and tags

Add these fields to your app and DB first (if you have not done so in the last lab exercise.

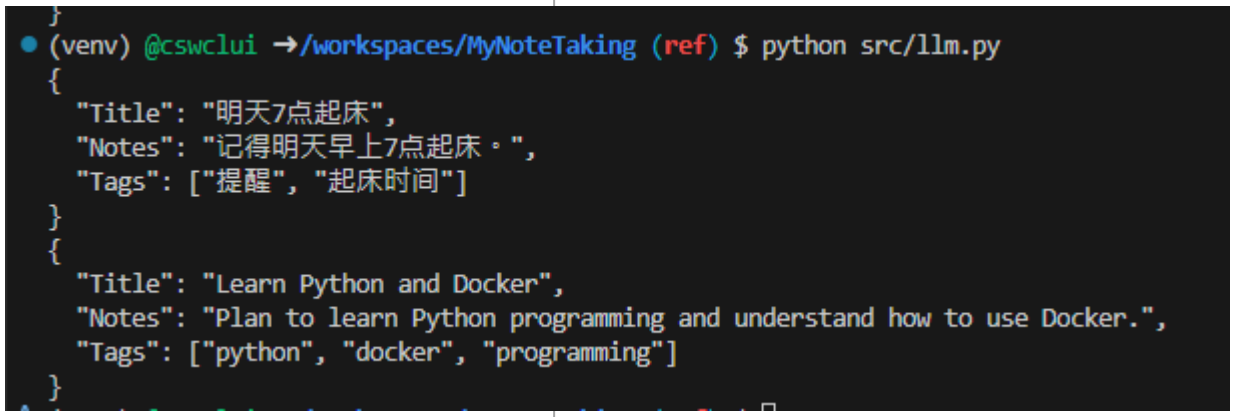
- First, we implement the helper function to generate the notes in structured JSON format.

```
system_prompt = '''
Extract the user's notes into the following structured fields:
1. Title: A concise title of the notes less than 5 words
2. Notes: The notes based on user input written in full sentences.
3. Tags (A list):  At most 3 Keywords or tags that categorize the content of the notes.


Output in JSON format without ```json. Output title and notes in the language: {lang}.
Example:
Input: "Badminton tmr 5pm @polyu".
Output:
{{
  "Title": "Badminton at PolyU",
  "Notes": "Remember to play badminton at 5pm tomorrow at PolyU.",
  "Tags": ["badminton", "sports"]
}}

'''
```

Alternatively, you may also store the system prompt in a file (to facilitating easy customization).

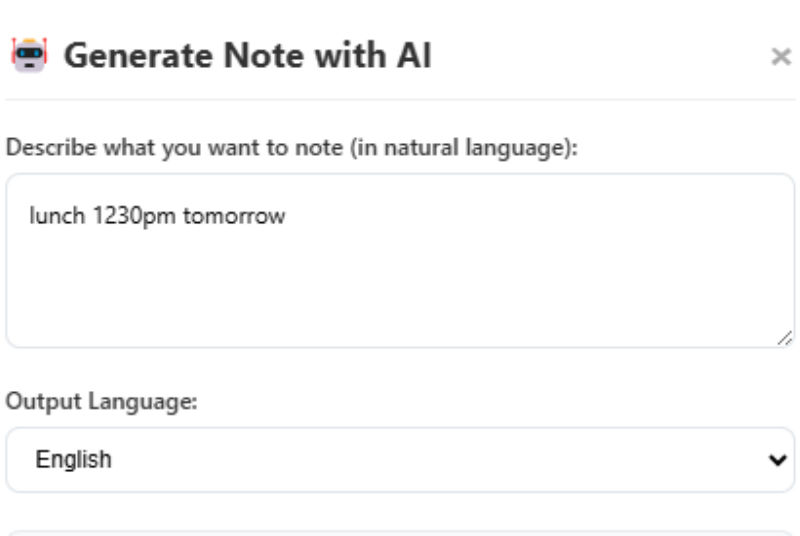
```python
def process_user_notes(language, user_input):
    system_prompt_filled = system_prompt.format(
        lang=language
    )

    messages = [
        {
            "role": "system",
            "content": system_prompt_filled,
        },
        {
            "role": "user",
            "content": user_input,
        }
    ]

    response_content = call_llm_model(model, messages)
    return response_content

# Run the main function if this script is executed
if __name__ == "__main__":
   result = process_user_notes("Chinese", "Get up tomorrow 7am")
   print(result)

   result = process_user_notes("English", "Learn python programming and docker")
   print(result)
```

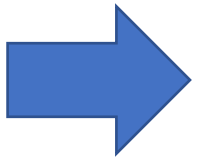
Run the program

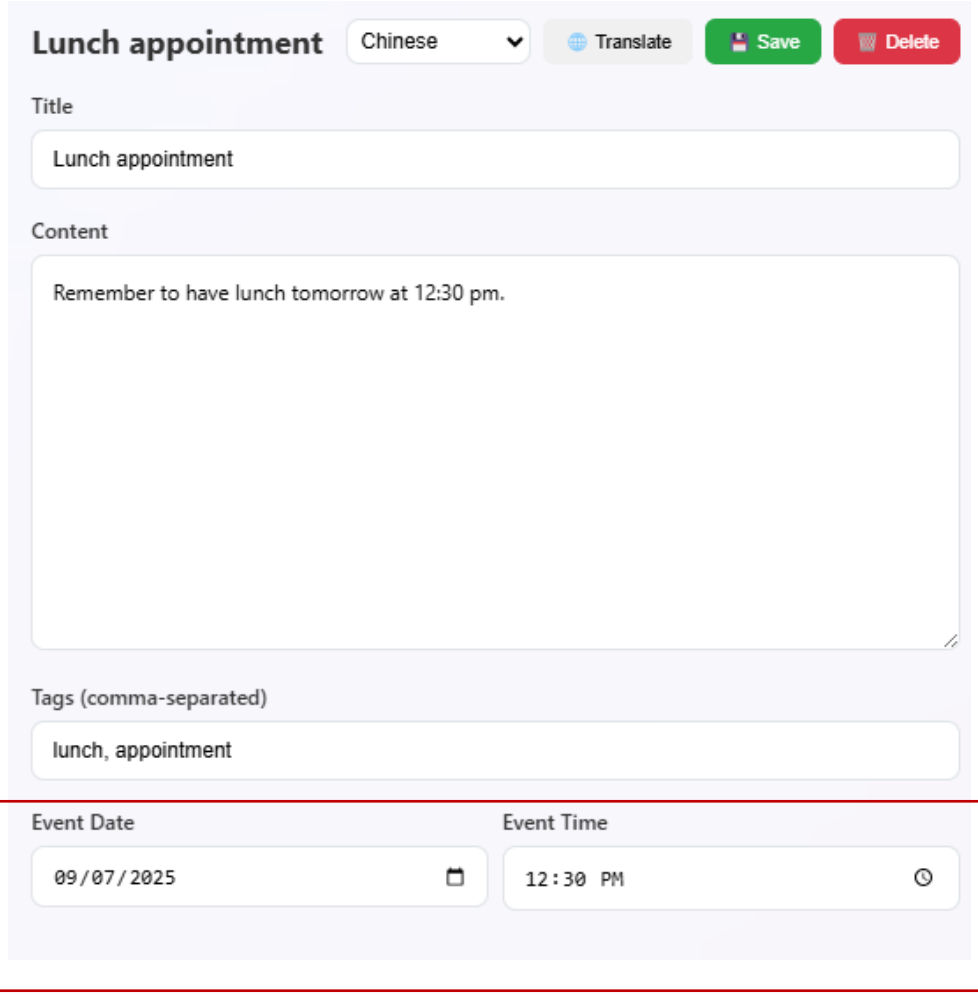


Use the agent mode (Claude 4) and ask AI to implement the feature by using the above helper function.

Exercise: Extend the “Generate notes” function to extract the date and time (if available)







You can first modify the helper function to extract the data and time from user input.
Then, use it as the context for the Copilot Agent to modify the app.

```
result = process_user_notes("Chinese", "Get up tomorrow 7am")
print(result)
```

```
(venv) @cswclui ➜/workspaces/MyNoteTaking (ref) $ python src/llm.py
{
  "Title": "明天早上7点起床",
  "Notes": "提醒自己明天早上7点起床。",
  "Tags": ["起床", "提醒"],
  "Date": "2025-09-07",
  "Time": "07:00"
}
```

LLM doesn’t have any knowledge about the current date/time.
How can we let the LLM knows the date for tomorrow?

Copilot instruction

- Defined in a markdown file named **.github/copilot-instructions.md**
  - Natural language guidance to help Github Copilot generate code and responses better tailored to a project's specific needs, team workflows, coding standards, and preferences.
  - E.g. You may define your coding style preference, how you want APIs called, naming patterns you want followed, etc
- The Copilot Agent will first read the file to understand the project contexts/rules before completing the actual task
- Purpose
  - Help align AI-generated code with human expectations, project standards, and team workflows.
  - Help the AI understand project-specific context, business logic, and domain knowledge

Example 1

```
This is a Next.js-based travel application with TypeScript that helps users search for
trips, manage bookings, view travel guides, and track points. The application uses
React components, server components, and client components as part of the Next.js App
Router architecture. Please follow these guidelines when contributing:

## Code Standards

### Required Before Each Commit
- Run `npm run lint` to ensure code follows project standards
- Make sure all components follow Next.js App Router patterns
- Client components should be marked with 'use client' when they use browser APIs or
React hooks
- When adding new functionality, make sure you update the README
- Make sure that the repository structure documentation is correct and accurate in the
Copilot Instructions file
- Ensure all tests pass by running `npm run test` in the terminal

### TypeScript and React Patterns
- Use TypeScript interfaces/types for all props and data structures
- Follow React best practices (hooks, functional components)
- Use proper state management techniques
- Components should be modular and follow single-responsibility principle

### Styling
- You must prioritize using Tailwind CSS classes as much as possible. If needed, you
may define custom Tailwind Classes / Styles. Creating custom CSS should be the last
approach.

## Development Flow
- Install dependencies: `npm install`
- Development server: `npm run dev`
- Build: `npm run build`
- Test: `npm run test`
- Lint: `npm run lint`
```

```
## Development Flow
- Install dependencies: `npm install`
- Development server: `npm run dev`
- Build: `npm run build`
- Test: `npm run test`
- Lint: `npm run lint`

## Repository Structure
- `app/`: Next.js App Router pages and layouts organized by route
- `components/`: Reusable React components
  - `components/ui/`: UI components (buttons, inputs, etc.)
  - `components/__tests__/`: Component tests
- `lib/`: Core logic and services
  - `lib/data/`: Data models and mock data
  - `lib/types/`: TypeScript type definitions
- `public/`: Static assets
- `tests/`: Test files and test utilities
- `README.md`: Project documentation

## Key Guidelines
1. Make sure to evaluate the components you're creating, and whether they need 'use
client'
2. Images should contain meaningful alt text unless they are purely for decoration. If
they are for decoration only, a null (empty) alt text should be provided (alt="") so
that the images are ignored by the screen reader.
3. Follow Next.js best practices for data fetching, routing, and rendering
4. Use proper error handling and loading states
5. Optimize components and pages for performance
```

https://github.blog/ai-and-ml/github-copilot/copilot-ask-edit-and-agent-modes-what-they-do-and-when-to-use-them

Example 2

Project Awareness & Context

- **Always read** `PLANNING.md` at the start of a new conversation to understand the project's architecture, goals, style, and constraints.
- **Check** `TASK.md` before starting a new task. If the task isn't listed, add it with a brief description and today's date.
- **Use consistent naming conventions, file structure, and architecture patterns** as described in `PLANNING.md`.
- **Use venv_linux** (the virtual environment) whenever executing Python commands, including for unit tests.

Code Structure & Modularity

- **Never create a file longer than 500 lines of code.** If a file approaches this limit, refactor by splitting it into modules or helper files.
- **Organize code into clearly separated modules**, grouped by feature or responsibility. For agents this looks like:
  - `agent.py` - Main agent definition and execution logic
  - `tools.py` - Tool functions used by the agent
  - `prompts.py` - System prompts
- **Use clear, consistent imports** (prefer relative imports within packages).
- **Use clear, consistent imports** (prefer relative imports within packages).
- **Use python_dotenv and load_env()** for environment variables.

Testing & Reliability

- **Always create Pytest unit tests for new features** (functions, classes, routes, etc).
- **After updating any logic**, check whether existing unit tests need to be updated. If so, do it.
- **Tests should live in a** `/tests` **folder** mirroring the main app structure.
  - Include at least:
    - 1 test for expected use
    - 1 edge case
    - 1 failure case

Task Completion

- **Mark completed tasks in** `TASK.md` immediately after finishing them.
- Add new sub-tasks or TODOs discovered during development to `TASK.md` under a "Discovered During Work" section.

Style & Conventions

- **Use Python** as the primary language.
- **Follow PEP8**, use type hints, and format with `black`.
- **Use** `pydantic` **for data validation.**
- Use `FastAPI` for APIs and `SQLAlchemy` or `SQLModel` for ORM if applicable.
- Write **docstrings for every function** using the Google style:

```
def example():
    """
    Brief summary.

    Args:
        param1 (type): Description.

    Returns:
        type: Description.
    """
```

Documentation & Explainability

- **Update** `README.md` when new features are added, dependencies change, or setup steps are modified.
- **Comment non-obvious code** and ensure everything is understandable to a mid-level developer.
- When writing complex logic, **add an inline** `# Reason:` **comment** explaining the why, not just the what.

AI Behavior Rules

- **Never assume missing context. Ask questions if uncertain.**
- **Never hallucinate libraries or functions** – only use known, verified Python packages.
- **Always confirm file paths and module names** exist before referencing them in code or tests.
- **Never delete or overwrite existing code** unless explicitly instructed to or if part of a task from `TASK.md`.

https://github.com/coleam00/context-engineering-intro/blob/main/CLAUDE.md